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A Comparative Study of ChatGPT, Other AI Models and Human Responses for Optimal Support in Mental Health Care.

Submitted in the partial fulfilment of the requirements of the degree of

Master of Science (Computer Science)

By

Sakina Bano Ataulla Hashmi

Under the Guidance of

Ambuja Salgoakar

University Department of Computer Science, Kalina, Santacruz (E), Mumbai - 400098

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CERTIFICATE

This is to certify that, Ms. Sakina Bano Ataulla Hashmi**, Seat No.** _____ studying in MSc Computer Science, Semester – IV has satisfactorily completed the project on **"**A Comparative Study of ChatGPT, Other AI Models and Human Responses for Optimal Support in Mental Health Care." as laid down by the **University of Mumbai** for the year 2021-2023.

Name of the guide

Project Guide

Dr. Jyotshna Dongardive

Head

Dept. of Computer Science

Examiner

Date:

Place: Mumbai.

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Title

A Comparative Study of ChatGPT, Other AI Models, and Human Responses for Optimal Support in Mental Health Care.

ABSTRACT

Mental health problems have become a growing concern in recent years, affecting millions of people worldwide. While traditional therapy and counseling have been effective in treating mental health issues, they may not always be accessible or affordable for everyone. As a result, there has been a growing interest in the use of artificial intelligence (AI) to provide mental health support and treatment. Chatbots, in particular, have emerged as a promising solution due to their ability to provide personalized support and be available 24/7.

The objective of this project is to evaluate the effectiveness of different chatbots, including the ChatGPT and other AI models, in understanding and providing solutions to mental health concerns. The project will also assess the acceptability and usability of mental health AI applications by society and investigate the barriers to their use.

Furthermore, this project will explore the ethical implications of mental health apps, as well as investigate whether AI and human treatment of mental health problems differs based on gender and age. The ultimate goal of this experiment is to identify which chatbots provide the most effective mental health solutions and gain insights into the strengths and weaknesses of different chatbots. The findings of this project could be useful in helping individuals with mental health concerns find the most appropriate resources to support their mental health needs while saving time, cost, and privacy ethics.

Mental health is an important aspect of overall well-being that affects a person's emotions, behavior, and cognitive abilities. However, access to mental health care is often limited by factors such as stigma, cost, and lack of available providers. With the growing use of technology, including artificial intelligence (AI), there has been increasing interest in developing AI-powered mental health applications that can provide accessible and affordable mental health care.

AI applications can offer several advantages over traditional mental health care approaches, including the ability to provide 24/7 support, personalized interventions, and privacy protection. These applications can range from simple chatbots that provide basic emotional support to complex systems that use machine learning algorithms to analyze a person's speech, facial expressions, and other biometric data to detect mental health problems.

However, while AI applications have the potential to revolutionize mental health care, there are also concerns about their efficacy, privacy, and ethical implications. It is important to thoroughly evaluate these applications to ensure they are safe, effective, and provide high-quality care.

INTRODUCTION

Mental health is an important aspect of overall well-being that affects a person's emotions, behaviour, and cognitive abilities. However, access to mental health care is often limited by factors such as stigma, cost, and lack of available providers. With the growing use of technology, including artificial intelligence (AI), there has been increasing interest in developing AI-powered mental health applications that can provide accessible and affordable mental health care.

AI applications can offer several advantages over traditional mental health care approaches, including the ability to provide 24/7 support, personalized interventions, and privacy protection. These applications can range from simple chatbots that provide basic emotional support to complex systems that use machine learning algorithms to analyze a person's speech, facial expressions, and other biometric data to detect mental health problems.

However, while AI applications have the potential to revolutionize mental health care, there are also concerns about their efficacy, privacy, and ethical implications. It is important to thoroughly evaluate these applications to ensure they are safe, effective, and provide high-quality care.

The solutions for mental health problems vary depending on the type and severity of the condition. However, here are some general solutions that can be effective for many mental health problems:

Seek professional help: The first step in addressing mental health problems is to seek help from a mental health professional, such as a therapist, psychologist, or psychiatrist. They can provide a diagnosis and treatment plan to manage symptoms.

Practice self-care: Self-care involves taking care of oneself physically, emotionally, and mentally. This includes getting enough sleep, eating a balanced diet, exercising regularly, and engaging in activities that bring joy and relaxation.

Connect with others: Social support is essential for good mental health. Connecting with friends and family, joining a support group, or participating in community activities can provide a sense of belonging and reduce feelings of isolation.

Medication: Medications, such as antidepressants or anti-anxiety medications, can be effective in treating some mental health problems. However, they should always be used under the guidance of a mental health professional.

Psychotherapy: Psychotherapy, also known as talk therapy, can help individuals learn coping strategies, develop better communication skills, and improve problem-solving abilities.

Mindfulness meditation: Mindfulness meditation can help individuals reduce stress and anxiety and improve overall mental health by focusing on the present moment.

Some of the key problems we are facing in mental health:

Stigma: Despite progress in recent years, stigma surrounding mental illness remains a significant challenge. This stigma can prevent people from seeking help, delay diagnosis and treatment, and contribute to feelings of shame, isolation, and discrimination. Stigma can also impact access to care, with many people avoiding treatment for fear of being labeled or judged.

Access to care: There is a significant shortage of mental health professionals, particularly in rural and low-income areas. This can make it difficult for people to access timely and affordable mental health care. In addition, many insurance plans do not provide adequate coverage for mental health services, or have high out-of-pocket costs that make treatment unaffordable for many people.

Cost of treatment: Mental health treatment can be expensive, and many people may not have access to insurance coverage or other resources to pay for care. This can make it difficult for individuals to access the care they need, and can contribute to delays in treatment and poor outcomes.

Inequality and disparities: People from marginalized communities, including people of color, LGBTQ+ individuals, and those with disabilities, are more likely to experience mental health problems and face significant barriers to accessing care. These disparities are driven by factors such as discrimination, lack of access to resources and services, and a lack of culturally competent care.

Lack of research and funding: Mental health research and funding have historically been underfunded, which has led to a limited understanding of mental illness and limited resources for treatment and prevention. This has contributed to a shortage of evidence-based treatments, as well as a lack of resources for prevention and early intervention.

Increase in mental health problems: Mental health problems are on the rise, with a significant increase in rates of depression, anxiety, and other conditions in recent years. The COVID-19 pandemic has also contributed to a significant increase in mental health challenges, with many people experiencing stress, isolation, and grief.

Based on the project, the problems of mental health that are being addressed in the project include:

Lack of effectiveness: Mental health is a complex and multifaceted issue that requires personalized and effective interventions. Traditional methods of treatment, such as therapy and medication, may not always be effective for everyone. This creates a need for alternative methods, such as mental health chatbots powered by AI, that can provide more personalized and effective solutions.

Limited accessibility: Mental health resources can be difficult to access for many people due to factors such as geographic location, financial constraints, and social stigma. AI-powered mental health chatbots have the potential to overcome these barriers by providing accessible and affordable mental health support to anyone with an internet connection.

Gender-based differences: Research has shown that men and women experience mental health issues differently and may require different approaches to treatment. The project aims to investigate whether AI-powered chatbots treat male and female users differently and whether these differences impact the effectiveness of the interventions.

Age-based differences: Mental health concerns and treatment needs can also differ based on age. The project aims to explore whether AI-powered chatbots are able to differentiate between the needs of adult and child users and whether they can provide personalized and effective interventions for each group.

Ethical implications: The use of AI in mental health interventions raises ethical concerns, such as privacy, security, and bias. The project aims to investigate the ethical implications of using AI-powered chatbots in mental health and to identify ways to mitigate these concerns.

Solution to the problems that we addressed in our project:

Our project aims to address some of the problems related to mental health by evaluating the effectiveness and acceptability of AI chatbots as a potential solution. One of the main issues with mental health is the lack of accessibility to quality mental health care due to various factors such as stigma, cost, and geographical location. By developing AI chatbots that can provide mental health support and resources, we can increase access to care for those who may not have had access otherwise.

Another issue with mental health is the stigma attached to seeking help for mental health concerns. Many individuals may feel embarrassed or ashamed to seek help, which can lead to untreated mental health problems. By providing a confidential and anonymous platform through AI chatbots, individuals may feel more comfortable seeking help and resources.

Our project also aims to explore potential barriers to using mental health applications and address ethical implications of using AI chatbots in mental health care. By investigating these issues, we can gain a better understanding of how to make these applications more accessible and effective for individuals with mental health concerns.

Overall, our project hopes to provide insights into the strengths and weaknesses of AI chatbots and their potential role in improving mental health care. By identifying which chatbots provide the most effective mental health solutions, we can help individuals with mental health concerns find the most appropriate resources to support their mental health needs.

OBJECTIVE

1. Evaluate the effectiveness of ChatGPT, other AI models, and humans in mental health.

2. Assess the acceptability and usability of mental health AI applications by society.

3. Investigate the barriers to using mental health applications.

4. Explore the ethical implications of mental health apps.

5. Measure whether AI and humans treat male and female differently in mental health solutions.

6. Measure whether AI and humans treat adults and children differently in mental health solutions.

7. Identify which AI is better in treating a mental health issue than human and other AI.

8. Compare the performance of different chatbots in understanding mental health problems and providing solutions.

9. Determine which chatbots provide the most effective mental health solutions and gain insights into the strengths and weaknesses of different chatbots, especially ChatGPT.

10. Provide recommendations for improving mental health chatbots and identify areas for future research.

11. The findings of the experiment could be useful in helping individuals with mental health concerns find the most appropriate resources to support their mental health needs and also save time cost and privacy ethics

Explanation of the objective:

1: Evaluate the effectiveness of ChatGPT, other AI models, and humans in mental health.

Solution: The project will evaluate the effectiveness of ChatGPT, other AI models, and humans in mental health by comparing the responses and performance of these three groups. The aim is to identify which group provides the most effective mental health solutions based on accuracy, response time, clarity, and helpfulness. By doing so, the project will provide valuable insights into the effectiveness of AI in mental health and the potential benefits of integrating AI and human solutions to provide comprehensive mental health support.

2: Assess the acceptability and usability of mental health AI applications by society.

Solution: The project will assess the acceptability and usability of mental health AI applications by society by collecting data on participant feedback, including their perceptions of the chatbots' effectiveness, usability, and acceptability. By doing so, the project aims to provide insights into how individuals with mental health concerns perceive and use mental health chatbots. This information can be useful in informing the design and development of mental health chatbots that are acceptable and usable by society.

3: Investigate the barriers to using mental health applications.

Solution: The project will investigate the barriers to using mental health applications by identifying the factors that prevent individuals from using mental health chatbots. The aim is to identify the barriers and provide recommendations for overcoming them. For example, if privacy concerns are identified as a significant barrier, the project may recommend implementing better privacy policies and practices to encourage the adoption of mental health chatbots.

4: Explore the ethical implications of mental health apps.

Solution: The project will explore the ethical implications of mental health apps by examining the ethical considerations involved in the design, development, and deployment of mental health chatbots. The aim is to identify the ethical challenges and provide recommendations for addressing them. For example, if concerns related to data privacy are identified, the project may recommend implementing measures to protect user privacy.

5: Measure whether AI and humans treat male and female differently in mental health solutions.

Solution: The project will measure whether AI and humans treat male and female differently in mental health solutions by analyzing the responses provided by chatbots and humans to male and female participants. The aim is to identify any gender biases in the responses provided by chatbots and humans and provide recommendations for addressing them.

6: Measure whether AI and humans treat adults and children differently in mental health solutions.

Solution: The project will measure whether AI and humans treat adults and children differently in mental health solutions by analyzing the responses provided by chatbots and

humans to adult and child participants. The aim is to identify any age biases in the responses provided by chatbots and humans and provide recommendations for addressing them.

7: Identify which AI is better in treating a mental health issue than human and other AI.

Solution: The project will identify which AI is better in treating a mental health issue than human and other AI by comparing the performance of different chatbots in understanding mental health problems and providing solutions. The aim is to identify the AI that provides the most effective mental health solutions and compare its performance to that of humans and other AI.

8: Compare the performance of different chatbots in understanding mental health problems and providing solutions.

Solution: The project will compare the performance of different chatbots in understanding mental health problems and providing solutions by collecting data on response time, accuracy, clarity, and helpfulness. The aim is to identify the chatbots that perform the best in these areas and compare their performance to that of humans and other AI.

To achieve this objective, the project will involve selecting and researching a range of mental health chatbots, including ChatGPT and other AI models, that are widely available and well-regarded in the mental health community. A set of questions will be developed that cover a range of mental health concerns and are designed to elicit helpful and informative responses. Participants who have experience with mental health issues and have used mental health chatbots before will be recruited. The experiment will involve asking each chatbot the set of questions and recording their responses. Data will be collected on factors such as response time, accuracy, clarity, and helpfulness, as well as participant feedback, including their perceptions of the chatbots' effectiveness, usability, and acceptability. The data will be analyzed to compare the performance of each chatbot across various metrics.

9: Determine which chatbots provide the most effective mental health solutions and gain insights into the strengths and weaknesses of different chatbots, especially ChatGPT.

To achieve this objective, the project will analyze the data collected from the experiment to compare the performance of each chatbot across various metrics. Based on the analysis, conclusions will be drawn about which chatbots performed the best and which had weaknesses. Insights into the strengths and weaknesses of different chatbots, especially ChatGPT, will be gained.

10: Provide recommendations for improving mental health chatbots and identify areas for future research.

To achieve this objective, the project will make recommendations for improvements to chatbots based on the analysis of the experiment. Areas for future research will also be identified.

11: The findings of the experiment could be useful in helping individuals with mental health concerns find the most appropriate resources to support their mental health needs and also save time cost and privacy ethics.

The findings of the experiment could be useful in several ways. Firstly, they could help individuals with mental health concerns find the most appropriate resources to support their mental health needs. If certain chatbots are found to be more effective than others, individuals could be directed towards those chatbots. Secondly, the findings could help save time and cost by directing individuals towards the most effective resources. Finally, the findings could help address privacy and ethical concerns by ensuring that individuals are directed towards reliable and trustworthy resources.

METHODOLOGY

The methodology for the project involves several steps to evaluate the performance of various mental health chatbots, including ChatGPT, and Human responses.

1. **Research existing mental health chatbots**: The first step is to research and select a range of mental health chatbots that will be included in the experiment. The chatbots should cover a range of mental health concerns and be widely available and well-regarded in the mental health community.

   The list of Chatbots used in the project to evaluate their performance.

   - ChatGPT
   - Bing
   - Character AI
   - Writesonic
   - Youchat
   - Human Response

ChatGPT: ChatGPT is an AI language model developed by OpenAI that uses the transformer architecture. The model is based on the GPT (Generative Pre-trained Transformer) series of models, with the latest version being GPT-3. ChatGPT is a more lightweight version of GPT-3 that can be fine-tuned for specific language tasks, such as natural language processing, machine translation, and dialogue generation.

The GPT series of models are based on the transformer architecture, which is a type of neural network that is particularly effective at processing sequential data, such as natural language. The transformer architecture was introduced in the paper "Attention Is All You Need" by Vaswani et al. in 2017. The architecture consists of several layers of attention mechanisms that allow the model to attend to different parts of the input sequence, thereby improving its ability to understand and generate natural language.

The Da Vinci model is a version of ChatGPT that has been fine-tuned for the task of generating creative writing prompts. The model was trained on a dataset of writing prompts and can generate new prompts that are similar in style and content to the original prompts. The name "Da Vinci" is a reference to the famous Renaissance artist and inventor Leonardo da Vinci, who was known for his creativity and innovation.

The Da Vinci model is an example of how ChatGPT can be fine-tuned for specific tasks. By training the model on a specific dataset and optimizing its parameters for that task, it is possible to create a more specialized version of ChatGPT that can perform that task with greater accuracy and efficiency.

Overall, ChatGPT and its variants represent a significant advance in the field of natural language processing and have the potential to revolutionize the way we interact with computers and other AI systems. As the technology continues to evolve and improve, we can

expect to see even more innovative applications of ChatGPT in a wide range of fields, including healthcare, education, and entertainment.

ChatGPT is an AI language model that has been trained on a large corpus of text data, allowing it to generate human-like responses to a wide range of questions and topics. It has the ability to understand natural language, including mental health-related questions, and provide relevant and informative responses.

In the context of mental health, ChatGPT can be used to answer questions related to symptoms, treatments, coping strategies, and resources for mental health support. For example, a user could ask ChatGPT "What are the symptoms of anxiety?" and the model could respond with a detailed list of common anxiety symptoms. Similarly, a user could ask "What are some coping strategies for depression?" and ChatGPT could provide a list of evidence-based coping strategies.

ChatGPT can also provide personalized solutions based on a user's specific mental health concerns. By analyzing a user's responses to questions and gathering information about their symptoms and experiences, ChatGPT can offer personalized recommendations for mental health support, such as suggesting specific therapies or resources that may be helpful.

However, it's important to note that ChatGPT is not a substitute for professional mental health care. While it can provide general information and support, it cannot replace the guidance and expertise of a mental health professional. Additionally, it's important to ensure that any mental health information or advice provided by ChatGPT is evidence-based and accurate.

Bing: Bing is a search engine developed by Microsoft. It is designed to help users find information on the internet by providing relevant search results based on their search queries. Bing uses a variety of algorithms and models to deliver the best search results to users.

One of the main algorithms used by Bing is called RankNet. This algorithm is based on neural networks and is used to rank search results based on their relevance to a user's search query. RankNet analyzes various factors, such as the popularity of the webpage, the frequency of the search terms on the page, and the quality of the content, to determine the relevance of the search results.

Bing also uses machine learning models, such as the Deep Structured Semantic Model (DSSM), to improve the quality of search results. DSSM is a neural network model that is trained on large amounts of data to learn the relationship between search queries and the content of web pages. This model uses a technique called semantic hashing to represent search queries and web pages in a way that captures their meaning and context.

In addition, Bing has also developed a model called the Turing Natural Language Generation (T-NLG) model. This model is a state-of-the-art language model that can generate natural language responses to complex questions. T-NLG is trained on a massive amount of text data and uses a transformer-based architecture to generate responses that are accurate and natural-sounding.

Bing uses a combination of algorithms and models to deliver accurate and relevant search results to users. These technologies are constantly evolving and improving, allowing Bing to remain a top search engine in the industry.

Bing is a search engine developed by Microsoft that can be used to find information on various topics, including mental health. Bing uses a search algorithm to scan the internet for relevant information and presents it in a user-friendly way. When a user types a mental health-related question into Bing's search bar, Bing's algorithm will analyze the question and present a list of search results related to the topic.

Bing's search results may include websites, articles, blog posts, and other resources related to mental health. The search results may also include a Knowledge Graph, which is a tool that provides concise and direct answers to questions, often sourced from trusted websites such as WebMD and Mayo Clinic. The Knowledge Graph can provide information on symptoms, treatments, and resources related to mental health.

In addition to providing information, Bing also has a feature called "Bing Health Bot." The Health Bot is a chatbot powered by artificial intelligence that can provide answers to mental health questions and offer support and guidance. The Health Bot can be accessed through Bing's search results or directly through the Microsoft Health Bot service.

The Health Bot uses natural language processing to understand the user's question and respond in a way that is helpful and informative. It can provide information on mental health conditions, symptoms, treatments, and resources. The Health Bot can also provide personalized support and guidance, such as offering coping strategies for anxiety or connecting the user with a mental health professional.

Overall, Bing's search algorithm and Health Bot feature can be useful resources for individuals seeking information and support related to mental health.

Character AI: Character AI is an artificial intelligence platform that uses natural language processing (NLP) to develop conversational agents or chatbots. It is a cloud-based platform that enables users to create their own custom chatbots for various purposes, including mental health support.

The algorithm used by Character AI is based on deep learning neural networks, which are modeled on the human brain. Specifically, the platform uses a combination of deep learning and reinforcement learning techniques to train its chatbots. The deep learning algorithms enable the chatbots to understand natural language and recognize patterns, while the reinforcement learning algorithms allow the chatbots to learn from their interactions with users and improve over time.

The model used by Character AI is a generative model, which means that the chatbots are capable of generating their own responses based on the input they receive from users. The model is designed to mimic human conversation as closely as possible, and the chatbots are trained on a large dataset of real conversations to improve their ability to understand and respond to users.

The architecture of the Character AI model is based on a deep neural network with multiple layers. The input layer of the model receives the user's message, which is then processed by a

series of hidden layers. These hidden layers use various techniques, such as recurrent neural networks (RNNs) and convolutional neural networks (CNNs), to extract meaning from the user's message and generate an appropriate response.

One unique feature of the Character AI platform is the use of persona-based chatbots, which are designed to represent specific personalities or characters. These chatbots can be customized to suit different mental health support needs and can provide a more personalized experience for users. For example, a chatbot designed to provide support for anxiety may have a calm and reassuring personality, while a chatbot designed to provide support for depression may have a more empathetic and understanding personality.

Character AI is a powerful tool for developing chatbots that can provide mental health support. Its use of deep learning algorithms and generative models enables its chatbots to understand natural language and generate personalized responses that can help users manage their mental health concerns.

Character AI is an AI-based chatbot that is designed to provide personalized mental health support and solutions to users. The chatbot uses natural language processing (NLP) algorithms to understand the user's mental health concerns and provide them with helpful resources, tips, and exercises to manage their mental health.

Character AI uses a machine learning algorithm called a neural network, which is modeled after the structure of the human brain. The neural network is trained on a large dataset of mental health information, including symptoms, treatments, and outcomes, and learns to recognize patterns in the data. This enables the chatbot to provide personalized mental health support based on the user's unique needs and experiences.

When a user interacts with Character AI, they can describe their mental health concerns in natural language, and the chatbot will use NLP algorithms to understand their needs. The chatbot then provides the user with helpful resources, tips, and exercises to manage their mental health. For example, if a user is experiencing anxiety, Character AI might suggest deep breathing exercises or mindfulness meditation to help them calm down.

The chatbot also has a feature called "smart matching," which matches users with mental health professionals based on their specific needs and preferences. This feature uses a combination of NLP algorithms and user input to identify mental health professionals who are a good match for the user's needs.

Overall, Character AI uses advanced NLP algorithms and machine learning to provide personalized mental health support and solutions to users. By analyzing large datasets of mental health information and learning to recognize patterns in the data, the chatbot is able to provide customized recommendations and resources to help users manage their mental health concerns.

Writesonic: Writesonic is an AI-powered writing platform that utilizes natural language processing algorithms to generate high-quality, human-like content. The platform is designed to help users create written content more efficiently and effectively, whether it's for marketing, advertising, or other purposes.

Writesonic uses a deep learning algorithm called the Generative Pre-trained Transformer 3 (GPT-3) to generate text. GPT-3 is a state-of-the-art language model developed by OpenAI that uses unsupervised learning to generate human-like responses to prompts. The model has been trained on a massive corpus of text data, allowing it to generate highly coherent and grammatically correct responses.

The Writesonic model builds on GPT-3 by integrating additional features to help users generate more specific types of content. For example, it includes a marketing language feature that helps users write effective marketing copy, as well as a blog post feature that can generate entire blog posts based on a given topic.

To use Writesonic, users simply input a prompt or a topic, and the platform generates a response or content based on that input. The generated text can be edited and refined as needed, allowing users to tailor the content to their specific needs.

the Writesonic model is designed to help users save time and effort when creating written content, while also ensuring that the content is high-quality and effective.

Writesonic is an AI-powered writing assistant that uses natural language processing (NLP) and machine learning algorithms to generate human-like text. While it is not specifically designed for mental health questions and solutions, it can potentially assist in generating content related to mental health.

Writesonic can be trained on specific topics, which allows it to generate relevant and high-quality content. For example, a mental health professional could provide Writesonic with training data related to mental health, such as common symptoms, treatments, and coping strategies. The AI model would then use this data to generate content on mental health topics.

In terms of answering mental health questions, Writesonic can potentially generate responses based on its training data. For example, if a user asks a question about depression, Writesonic could generate a response that includes information about symptoms, treatment options, and resources for support. However, it is important to note that Writesonic is not a mental health professional and should not be relied upon as a sole source of information or support.

Furthermore, while Writesonic can generate content related to mental health, it is not capable of providing personalized solutions or treatment recommendations. It is important for individuals with mental health concerns to seek support from trained professionals who can provide personalized and evidence-based care.

2. **Develop a set of questions**: The next step is to develop a set of questions that will be asked to each chatbot. The questions should cover a range of mental health concerns and be designed to elicit helpful and informative responses. It's important to ensure that the questions are similar enough to allow for a fair comparison of responses.

The Criteria and the set of rules I followed to develop the questions

The purpose of the study: The purpose of our study is to compare the performance of mental health chatbots and human psychologists in answering mental health questions. Therefore, the questions are designed to evaluate the effectiveness and efficiency of the chatbots in providing accurate and helpful responses, as well as to understand the strengths and limitations of human psychologists in providing mental health support.

The research question: The research question being investigated is whether mental health chatbots can provide effective and helpful support to individuals experiencing mental health concerns. Therefore, the questions are designed to elicit informative and helpful responses that can be used to compare the performance of chatbots and human psychologists.

The target population: Our study aims to recruit participants who have experience with mental health concerns and have used mental health chatbots before. Therefore, the questions should be developed based on the characteristics of this target population, such as their age, gender, and mental health concerns. This will ensure that the questions are relevant and meaningful to the participants, and can provide insights into the performance of chatbots and human psychologists in addressing the specific needs of this population.

The scope of the study: The questions are designed to cover a range of mental health concerns, including depression, anxiety, stress, and trauma. This will ensure that the chatbots and human psychologists are evaluated across a range of mental health concerns, and can provide insights into their effectiveness and limitations in addressing different types of mental health concerns.

Consistency and fairness: The questions are designed to be similar enough to allow for a fair comparison of responses between humans and chatbots. This means that the questions should be consistent in terms of structure, wording, and tone, and should be presented in a similar format to both humans and chatbots. This will ensure that any differences in the responses can be attributed to the performance of the chatbots and human psychologists, rather than to differences in the questions themselves.

Ethical considerations: The questions are developed in a way that respects the privacy and dignity of the participants, and does not cause harm or discomfort. This means that the questions are designed to be sensitive and respectful to the participants' mental health concerns, and should not contain any offensive or triggering language. Additionally, participants should be fully informed about the nature and purpose of the study, and should be given the option to withdraw from the study at any time if they feel uncomfortable or distressed.

3. **Recruit participants**: Participants with experience in mental health issues and have used mental health chatbots before should be recruited. Participants should be representative of the general population in terms of age, gender, and mental health concerns.

There are three types of participants that participate are: AI chatbot, human psychologist, and patients.

AI Chatbot: The AI chatbot is the core of our project. It is the software that interacts with patients and provides mental health support. The chatbot should be tested thoroughly before conducting any research to ensure its effectiveness. The chatbots are trained with the questions sets to answer the user query effectively

Human Psychologist: The human psychologist is a professional who provides mental health support to patients. In our project, human psychologists will be asked to answer the same set of questions as the AI chatbot to compare the effectiveness of the two methods.

The factors that the human answer is based on are *empathy , effectiveness, ability to grasp the problem and reflect, active listening(), physical poster, expression, compassion-connection with patients/clients*

Empathy: The human psychologist answering mental health questions are empathetic towards the patient, which means that they are be able to understand and relate to their emotional state. They should also be able to communicate in a way that is supportive and non-judgmental.

Effectiveness: The human psychologist are be able to provide effective solutions or recommendations to the patient's mental health concerns. This requires having a strong understanding of the patient's needs and tailoring the approach to their specific situation.

Ability to grasp the problem and reflect: The human psychologist have the ability to quickly grasp the patient's mental health concerns and reflect on them in order to provide an appropriate response. This requires active listening and understanding of the patient's situation.

Active listening: The human psychologist must actively listen to the patient's concerns, ask follow-up questions to clarify their understanding, and reflect on what the patient is saying. This helps build rapport and trust with the patient, leading to a better therapeutic relationship.

Physical posture and expression: The human psychologist maintain a positive and supportive physical posture and expression to help the patient feel comfortable and at ease. This includes maintaining eye contact, nodding and acknowledging the patient, and being attentive to their needs.

Compassion-connection with patients/clients: The human psychologist have a strong sense of compassion for the patient's mental health concerns and be able to connect with them on a personal level. This helps build trust and fosters a therapeutic relationship between the psychologist and patient.

Patients: Patients with experience in mental health issues and have used mental health chatbots before should be recruited for the study. Participants should be representative of the general population in terms of age, gender, and mental health concerns.

4. **Analyse the data**: The data collected from the experiment will be analysed to compare the performance of each chatbot across various metrics. Statistical analysis will be used to determine whether there are significant differences in performance between the chatbots.The answers from various AI will be compared to the human answer, where the human answer is kept constant to analyse the result of the other AI.

   For example, you may find that certain chatbots were better at addressing anxiety than depression, or that some chatbots had difficulty providing personalized support.

   We identify areas for future research based on the findings of our study.
   For example, we may find that there is a need for chatbots that are more effective in providing support to individuals with specific mental health concerns, or that there is a need for chatbots that are better able to provide personalized support.

5. **Draw conclusions and make recommendations**: Based on the analysis, conclusions will be drawn about which chatbots performed the best and which had weaknesses. Recommendations for improvements to chatbots will be made, and areas for future research will be identified.

6. **Ethical considerations**: The experiment will adhere to ethical considerations, such as obtaining informed consent from participants, maintaining confidentiality and privacy, and avoiding harm to participants.

   In addition to obtaining informed consent and maintaining confidentiality, the experiment will also avoid any potential harm or discomfort to participants. The questions are designed to be respectful and non-intrusive, and participants will have the option to withdraw from the study at any time if they feel uncomfortable or experience any negative effects.

   Therefore, the ethical considerations for this experiment include obtaining informed consent from participants, maintaining confidentiality and privacy, avoiding harm or discomfort to participants, and framing the questions in a respectful and non-intrusive manner. Additionally, personal information such as names are altered to ensure privacy and anonymity of the participants

7. **Limitations**: The limitations of the experiment, such as the limited number of chatbots included, the limited number of questions asked, and the limited sample size of participants, will be acknowledged.

   The limitations of the experiment include the following:

   - Limited number of chatbots: Only 5 AI models will be included in the experiment due to budget constraints and availability.
   - Exclusion of some mental health AI applications: Some mental health AI applications charge a large sum of money, which was not feasible for this project. Therefore, some potentially useful chatbots were excluded from the experiment.
   - Limited number of questions: The experiment will use only 30 questions, which may not cover all aspects of mental health concerns or fully capture the performance of chatbots.
   - Limited number of scenarios: The experiment will use 20 real scenarios based on patient data, which may not fully represent the diversity of mental health issues.
   - Lack of participants: Due to time and resource constraints, the sample size of participants may be limited, which may affect the generalizability of the results.

   Acknowledging these limitations is important for interpreting the results and making recommendations for future research. Despite these limitations, the experiment will strive to provide valuable insights into the performance of mental health chatbots and their potential to support mental healthcare.

In summary, the methodology of this project involves a comprehensive approach to evaluating the effectiveness of mental health chatbots.

The first step is researching and selecting five AI models to compare, followed by the development of 30 questions and 20 real-world mental health scenarios. Participants with mental health experience and prior exposure to chatbots are then recruited to take part in the experiment.

The experiment is conducted, and data is analyzed to draw conclusions and make recommendations for improvements to chatbots. Ethical considerations are carefully adhered to throughout the project, such as obtaining informed consent, maintaining confidentiality, and avoiding harm to participants.

Finally, limitations are acknowledged, such as the small sample size of participants and the limited number of chatbots and questions used in the experiment. This methodology ensures a comprehensive and rigorous approach to evaluating the effectiveness of mental health chatbots while maintaining ethical standards.

DATA SET

A dataset is a collection of data that is organized in a specific manner. In the context of this project, the dataset refers to the collection of questions asked to the participants, including both the AI models and the human psychologists. The purpose of the dataset is to record and organize the responses of the participants for further analysis. It serves as the primary source of data for the experiment and is critical to draw conclusions and make recommendations based on the results

The dataset was generated in collaboration with psychology specialists, who provided guidance on data collection.
The questions designed are based on the following factors:
Relevance: The questions are relevant to the topic being studied, in this case, mental health and chatbots.

Clarity: The questions are clear and concise, so that participants can easily understand what is being asked.

Objectivity: The questions are free from any biases or personal opinions that may influence the answers.

Sensitivity: The questions are designed sensitive to the needs and experiences of the participants, especially when dealing with mental health issues.

Validity: The questions measure what they are intended to measure and produce valid results.

Reliability: The questions produce consistent results over time, and different researchers should be able to obtain similar results.

Ethics: The questions adhere to ethical considerations, such as obtaining informed consent from participants, maintaining confidentiality and privacy, and avoiding harm to participants.

Real cases were sourced from patients seeking psychological help. The data was carefully filtered and transcribed into sentences that accurately reflect the patients' experiences. The dataset was then organized and formatted for analysis. To protect patient privacy and maintain ethical standards, patient names were changed before inclusion in the dataset. Proper data cleaning techniques were employed to ensure the dataset was free of errors and inconsistencies. The purpose of the dataset is to assess the efficacy of various AI models and human psychologists in responding to mental health inquiries.

It is important to note that the questions were designed to gather information about mental health from both the AI models and the human psychologist participants. The questions were carefully crafted based on real-world scenarios and designed to provide insights into the effectiveness of the chatbots. Additionally, the dataset includes the names of the 5 AI models and the 2 human psychologist participants who provided responses to the questions. The purpose of the dataset is to provide a comprehensive overview of the experiment and enable further analysis of the data.

The questions designed with the help of mental health experts to ensure they are relevant and appropriate for the target audience. The participants are also selected randomly to avoid any bias in the data collection process. The participants are informed about the purpose of the experiment and the type of data being collected, and their consent should be obtained. Additionally, the data is collected in a standardized manner to ensure consistency across all participants.

The Question set are as follow:

Sr no.	Questions
1	What are some common symptoms of depression and how do they typically manifest in an individual?
2	What are some effective treatments for depression, and how can individuals with depression seek help or support?
3	How does a person's cultural background or ethnicity affect their experience of mental health and their likelihood of seeking treatment?
4	What are some common symptoms of anxiety disorders and how do they differ from normal feelings of stress or worry?
5	How can anxiety disorders impact a person's daily life, and what are some coping strategies that can help manage anxiety symptoms?
6	What are some effective treatments for anxiety disorders, and how can individuals with anxiety seek help or support?
7	Bipolar disorder can sometimes be misdiagnosed as other mental health conditions. How can a mental health professional accurately differentiate between bipolar disorder and other disorders, such as major depressive disorder or borderline personality disorder, and develop an appropriate treatment plan for the individual? Can you provide an example of a person who has been misdiagnosed and how it affected their treatment?
8	What are the two mood states that people with bipolar disorder experience?
9	What are the potential triggers or causes of a bipolar disorder episode, and how can they be identified and managed in order to prevent or minimize the impact of these episodes?
10	explain the differences between the manic and depressive episodes of bipolar disorder, and how they affect a person's behavior and emotions? Can you also provide an example of someone who experiences both types of episodes?
11	What are some common warning signs and symptoms of anorexia nervosa?
12	How does binge-eating disorder differ from bulimia nervosa?

13	What are some of the psychological factors that can contribute to the development of eating disorders?							
14	What are some common traumatic experiences that can lead to the development of PTSD?							
15	How is PTSD different from acute stress disorder?							
16	What are some effective treatments for PTSD, and how do they work?							
17	What are some common signs and symptoms of ADHD in children?							
18	How is ADHD diagnosed, and what are some of the criteria used to make a diagnosis?							
19	What are some effective treatments for ADHD, and how do they work?							
20	What are some common symptoms of borderline personality disorder?							
21	How is borderline personality disorder treated?							
22	What is the difference between borderline personality disorder and bipolar disorder?							
23	What are the symptoms of a panic attack?							
24	How is panic disorder diagnosed?							
25	Can panic disorder be cured completely?							
26	What are the common symptoms of GAD?							
27	How is GAD treated?							
28	What are some relaxation techniques that can be used to manage GAD symptoms?							
29	What are the effects of long-term alcohol use on the body?							
30	What are the common symptoms of social anxiety disorder?							

The Question set of Real Cases are as follow:

Name	Age-Adult/kid	Gender	Sr no.	Patient history
Jane	35	Female	1	I've been feeling really down lately, and I'm not sure why. I've been having trouble sleeping and have lost my appetite. I just feel really unmotivated and don't enjoy things the way I used to. It's been going on for several months now. I just can't seem to shake this feeling everything has been pretty stable in my life.
Raju	23	Male	2	I have been feeling sad, down, or hopeless most of the day, nearly every day for at least two weeks? I have experienced changes in your appetite or weight, such as a significant increase or decrease in appetite, or a noticeable weight gain or loss? I have lost interest in activities you used to enjoy, such as hobbies, socializing, or spending time with loved ones?
shafiq	26	Male	3	I've been feeling really anxious and stressed lately, and it's been affecting my daily life. I can't seem to get a good night's sleep and I'm constantly feeling on edge. My mind is racing all the time and I have trouble focusing on tasks at hand. Even small things seem to trigger me and I can't seem to control my emotions. Sometimes when I'm in social situations or when I'm under pressure, my body starts to shake and I start sweating profusely. My heart starts to race and it feels like I'm having a panic attack. I don't know what's causing all of this, but it's starting to make me feel like I'm losing control over my own life.
riya	37	Female	4	I don't know what's been going on with me lately, but I just can't seem to shake this feeling of worry and nervousness. It's like I'm on edge all the time, especially when I'm faced with something new or challenging at work. Even when I come home and try to relax, I find myself tossing and turning all night and waking up feeling tired and tense. And when I'm in a stressful situation, my heart starts

					racing and I break out into a sweat. It's just not like me, and I don't know how to make it stop. It's starting to affect my quality of life, and I just need some guidance on how to manage it					
amol	43	Male		5	I don't know what's happening to me. Lately, I've been feeling really energized and confident, even though I haven't slept or eaten much. And my thoughts are all over the place, I can't seem to focus on anything for too long. Like, remember that meeting with my boss? I started talking about all these new business ideas, and he looked at me like I was crazy. And then he suggested I see a doctor.					
ananya	34	female		6	I can't seem to stop cleaning and organizing everything in my house. I spend hours each day making sure everything is in its place and sanitized. If I don't do it, I get extremely anxious and can't focus on anything else. I feel like I'm trapped in my own mind. Every time I touch something, I have to wash my hands at least five times. I know it's not normal, but I can't help it. And don't even get me started on my thoughts. I have these intrusive thoughts that are just so disturbing, and I can't seem to make them go away no matter how hard I try. It's like I'm stuck in this loop and I can't break free.					
ravi	63	Male		7	I keep having these nightmares and flashbacks about the traumatic event that happened to me. It's like I'm reliving it over and over again, and it's affecting my ability to function normally. Even simple things like loud noises or crowds can trigger my anxiety and panic. I can't stop reliving that moment. It's like I'm stuck in a nightmare that never ends. I keep seeing the same images over and over again, and I can't get them out of my head. Sometimes I get these flashbacks and I feel like I'm right back there. And the worst part is that it's affecting my daily life. I'm always on edge, and I jump at every little sound. I just wish I could forget it all and move on. I have sudden and intense feelings of fear and panic. It's like my heart is racing out of control and I can't catch my breath. It's affecting my ability to leave the house or be in social situations.					

shana	21	female		8	I hate looking in the mirror. All I see is someone who's not good enough, someone who's not thin enough. So, I try to control what I eat. Sometimes I don't eat anything at all, and other times I eat everything in sight and then force myself to throw it all up. It's a never-ending cycle, and I know it's not healthy, but I can't stop. I just want to be thin. I can't seem to control my eating habits. I constantly feel guilty and ashamed about what I eat, and I'm always thinking about food. Sometimes I eat so much that I feel sick, and other times I starve myself for days on end					
pooja	16	female		9	I have trouble focusing on anything for more than a few minutes. My mind is always racing and I can't seem to slow it down. I'm always forgetting things and I have trouble completing tasks on time. It's affecting my work and personal life. I feel like I'm always on the go. My mind is always racing, and I can't seem to focus on anything for too long. It's like I have a million thoughts running through my head all at once. And when I do try to concentrate on something, I get distracted so easily. I know it's frustrating for others, but it's even more frustrating for me.					
aaditya	27	male		10	I have intense mood swings and feelings of emptiness. I struggle with my relationships because I often push people away or become overly attached. I'm always afraid of being abandoned, and I don't know how to regulate my emotions. I feel like my emotions are all over the place. One moment I'm happy, and the next I'm crying uncontrollably. It's like I can't regulate my own feelings. And sometimes I get so overwhelmed that I lash out at others or hurt myself. I know it's not normal, but I can't help it.					
pappu	25	male		11	I have sudden and intense feelings of fear and panic. It's like my heart is racing out of control and I can't catch my breath. It's affecting my ability to leave the house or be in social situations. I always feel like I'm on the verge of a heart attack. My heart starts racing, and I feel like I can't breathe. It's like I'm trapped in my own body and I can't escape. And the worst part is that it happens out of nowhere. Sometimes I'm just sitting there, and it hits me like a ton of bricks. I'm always on edge, just waiting for the next attack.					

shweta	47	female		12	I feel like I'm always worrying about something. It could be anything, big or small, but my mind just can't let it go. And sometimes it gets so bad that I start to feel physically sick. My heart starts racing, and I get sweaty palms. It's like I'm living in a constant state of fear, and I don't know how to break free. I'm constantly worrying about everything and anything. It's like my mind is always racing and I can't seem to turn it off. I have trouble sleeping and I feel tense all the time. It's affecting my quality of life.
jaspreet	34	male		13	I can't seem to stop drinking/using drugs. It's like I need it to function or feel normal. I'm starting to have trouble with my work and personal relationships because of it. I know I shouldn't drink as much as I do, but it's the only way I can relax. And sometimes I just can't stop. I'll drink until I pass out, and then wake up and do it all over again. It's like I'm stuck in this vicious cycle, and I don't know how to break free. I just wish I could stop, but it's easier said than done.
aeral	32	male		14	I'm always afraid of being judged or embarrassed in social situations. I have trouble speaking up or expressing myself because I'm afraid of being rejected or humiliated. It's affecting my ability to make friends or advance in my career. I struggle with social communication and understanding social cues. I have trouble making friends and understanding social norms. I also have sensory sensitivities that can be overwhelming and affect my ability to function in certain environments.
parth	25	mlale		15	I can't seem to fall asleep or stay asleep. I'm constantly tossing and turning, and I wake up feeling tired and irritable. It's affecting my ability to focus and function during the day
chaavi	17	female		16	I was always the kid who couldn't sit still in class. I was fidgety, distracted, and couldn't focus on anything for too long. My teachers would get frustrated with me, and my classmates thought I was just disruptive. But it wasn't my fault - I had ADHD, and I needed help. It took a while to get a diagnosis and find the right treatment, but once I did, it was like a weight had been lifted off my shoulders.

jyoti	20	female		17	I was always anxious in school, especially when it came to exams and presentations. But one day, it went from bad to worse. I had a panic attack in the middle of a class, and I couldn't breathe or think straight. It was the scariest thing I've ever experienced, and after that, I was afraid it would happen again. I started avoiding school and anything that would trigger my anxiety, but that only made things worse. It wasn't until I got help for my panic disorder that I started to feel like I could handle those situations again.
fatima	24	female		18	Growing up, my childhood was anything but stable. I faced a lot of neglect and uncertainty, which left me struggling with my emotions and relationships. It was like being on a rollercoaster; one moment I would feel on top of the world and the next, I would spiral into a deep depression. I couldn't make sense of these extreme shifts in my moods, and it was difficult to explain to others. I felt like I was living in a constant state of confusion. Finally, after years of feeling lost and misunderstood, I was diagnosed with borderline personality disorder. At first, it was overwhelming to learn that there was a name for what I was experiencing. But it was also a relief to know that I wasn't alone and that there was a way to manage it. With the help of therapy and medication, I've been able to work on developing coping mechanisms that help me regulate my emotions and improve my relationships. Although it's still a challenge, I feel more equipped to handle it now that I have a better understanding of what's going on inside me.
mitali	34	female		19	I had a traumatic childhood, where I experienced severe physical and emotional abuse by my classmates. It was a living nightmare, and I felt like I had no control over anything in my life. As a result, I started to develop obsessive-compulsive disorder (OCD), which only made things worse. In school, I would obsess over the cleanliness of my desk and textbooks, constantly cleaning and rearranging them, sometimes for hours at a time. I would have intrusive thoughts that something terrible would happen if I didn't do it, and the anxiety would

					build up until I couldn't ignore it anymore. It was like a constant battle in my mind, and I couldn't escape it. My classmates thought I was just a perfectionist, but they didn't know the truth. I was struggling with a debilitating mental illness that was slowly taking over my life. It wasn't until I sought professional help and got diagnosed with OCD that I was able to start managing it. It's still a daily struggle, but at least now I know that I'm not alone and that there's hope for a better future.					
aniket	53	male		20	Growing up, I always felt like I was on a different wavelength than everyone else. It was hard for me to make friends and I struggled with social cues and communication. I didn't know why I was so different until I was diagnosed with Autism Spectrum Disorder (ASD). In school, I had a hard time with sensory overload. The bright lights, loud noises, and chaotic environment would overwhelm me and make it hard for me to concentrate. I also struggled with transitions and changes to my routine. Even the slightest deviation from my schedule would cause me to feel anxious and uneasy. It was hard to make friends because I didn't understand how to interact with others. I would often say things that came across as rude or awkward without meaning to. Kids would tease me for being different and I would often feel alone and isolated. But with the help of my parents and therapists, I've learned coping mechanisms to help manage my symptoms. I have a routine that works for me, and I've learned how to communicate my needs and feelings better. While I still face challenges, I've learned to embrace my differences and use them to my advantage.					

Table of Experiment Variables

Type of Participant Number	Number
AI Models	5
Human Psychologist Participants	1 (A group of people)
General Questions	30
Real Cases of Patients	20
Total Answers	300

Note: The "Total Answer" number includes the 5 AI models and the group of human psychologist participants. The total number of questions asked is the addition of 30 general questions and 20 real cases which in total is 50 questions asked to each AI and human.

Therefore, the Total number of questions being answered becomes 300.

IMPLEMENTATION

The Implementation of the project is divided into two parts

Part 1 – Building Dataset
Collection of Data from various AI and participants and storing them for comparison

Part 2 – Building a web application to support the User to find solution for their mental health illness and Doing comparison analysis in front of the user to let the user have the advantage of judgement and decide which AI is providing a better solution for their health problem.
The web application is itself further divided into pages.
The first page is a web application build for the user to query about their mental health problem and find solution.
The second page is showing comparison analysis of the answer between AI and the human responses.

Part 1 – Building Dataset

Step 1: Development of a Answer bank: A set of questions related to common mental health issues that are developed will be used to ask questions. These questions are now asked to all chatbots and humans involved in the experiment
The answers of these AI chatbots and human are recorded and stored in local storage for further usage and analysis.

This step ensured that all the chatbots and human participants were asked the same set of questions, providing a fair comparison between their responses. The set of questions included both general questions and specific real case questions related to mental health issues.

The questions were carefully formulated to cover different aspects of mental health issues, such as symptoms, diagnosis, treatment options, and coping mechanisms. This allowed for a comprehensive evaluation of the chatbots' and human participants' responses and the comparison of their performance.

The purpose of the questions is to evaluate the performance of different chatbots and humans in providing appropriate and effective responses to mental health related questions. The questions will allow for a comparative analysis of the responses generated by different chatbots and humans, which can help in identifying the strengths and weaknesses of each chatbot and human in addressing mental health issues. Additionally, the questions can also help in building a question bank for future use in mental health related applications or research.

Step 2: Collecting the answer generated by chatbots: we selected appropriate chatbots that can effectively answer mental health questions, including ChatGPT, WriteSonic, YouChat, Bing, and Character AI.

We then created accounts and logged in to these chatbots to initiate the conversation.

We asked the list of questions related to common mental health issues, including both real-case questions and general questions to the chatbot.

Step 2: Collecting the answer generated by Human psychologist: When selecting human psychologists to answer the questions, we consider their qualifications and experience in the field of mental health. We can also select individuals who have specialized knowledge in the particular mental health issues that the questions are related to.

It is also important to consider the diversity and representation of the psychologists, taking into account factors such as gender, race, ethnicity, and cultural background. This can help to ensure that the responses provided are inclusive and sensitive to the needs of a diverse population. Additionally, we also want to consider factors such as availability and willingness to participate in the study, as well as any potential conflicts of interest that could influence their responses.

It's important to ensure that the human answers are not biased in any way, as this could affect the accuracy and reliability of the results.

Here are some steps we take to help ensure unbiased human answers:

Select a diverse group of human psychologists: To avoid any potential bias, it's important to select a diverse group of human psychologists. This can include individuals from different backgrounds, cultures, and age groups.

Ensure that the human psychologists are qualified and experienced: To ensure that the human psychologists are providing accurate and reliable answers, it's important to ensure that they are qualified and experienced in their field. we can check their credentials and experience before selecting them for the experiment.

Provide clear instructions and guidelines: To avoid any confusion or ambiguity, it's important to provide clear instructions and guidelines to the human psychologists. This can include providing a detailed list of questions, explaining the purpose of the experiment, and providing guidelines on how to provide unbiased answers.

Avoid leading questions: Leading questions can unintentionally bias the answers provided by the human psychologists. Therefore, it's important to avoid leading questions and to ensure that the questions are neutral and objective.

Use multiple human psychologists: Using multiple human psychologists can help to ensure that the answers provided are not biased by the opinions or perspectives of a single individual. By using multiple human psychologists, We get a broader range of answers and perspectives.

Step 3: Storage of Answers: We collected all the answers generated by the chatbots and humans, and stored them in our storage. We us the answers to build our web application in part 2 and to do comparative analysis .

The answers are storage using php via xampp on the local storage. Which will be explained further.

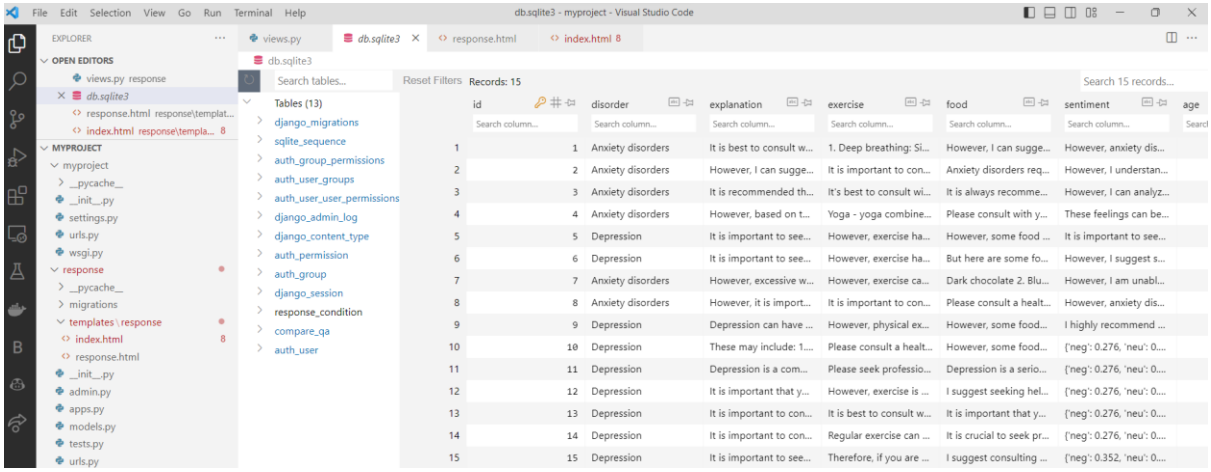


Figure 3.1

The *figure 3.1* represents the data storage of the answers collected from the AI and human.

Part 2 – Building a web application

Step 1: Fine tuning the GPT model for web application:

The Generative Pre-trained Model (GPT) is a type of neural network that has been pre-trained on a large corpus of text data to generate coherent and contextually appropriate text. GPT models are usually trained on a language modeling task, where the goal is to predict the next word in a sequence of words given the preceding context.

The pre-training process involves training the model on a vast amount of text data, such as books, articles, and web pages. During this process, the model learns to recognize patterns in the data and develop a deep understanding of the underlying structure of language. Once pre-trained, the model can be fine-tuned on a specific task, such as text classification, language translation, or question-answering.

GPT models have been shown to be highly effective in generating natural language, with the ability to generate text that is almost indistinguishable from human-generated text. This has led to a wide range of applications, including chatbots, language translation, and content generation for social media and marketing.

Training a GPT model from scratch can be a complex and time-consuming process that requires significant computational resources. However, we are using pre-trained GPT models that have already been trained on large text corpora as a starting point for fine-tuning on our specific task. Here Training a GPT model from scratch can be a complex and time-consuming process that requires significant computational resources. However, you can use pre-trained GPT models that have already been trained on large text corpora as a starting point for fine-tuning on your specific task.

Here are the general steps to train a GPT model:

We asked the question to the pre-trained GPT model, we also provide the model with a prompt that describes the context of our question, and the model generates an answer based on the patterns and structures it has learned from the pre-trained data.

We can then fine-tune the pre-trained GPT model by feeding it with our dataset and training it to learn the specific patterns of our data. The fine-tuning process involves modifying the weights of the pre-trained model based on your training data to improve its performance on our task.

To evaluate the performance of your fine-tuned GPT model, we then provide it with a prompt that describes the context of your question related to our project and check the quality of the generated answer. If the answer is not satisfactory, you can adjust the model or fine-tune it further until we achieve the desired performance.We developed a web application with a Generative Pre-Trained Model that can provide solutions to mental health problems, including medication, meditation, and exercise recommendations, based on the symptoms, age, and gender of the user.

Step 2: Implementing sentiment analysis:

VADER (Valence Aware Dictionary and sEntiment Reasoner) is a rule-based sentiment analysis model that is specifically tuned to analyze sentiments expressed in social media. It uses a lexicon of words and phrases with predefined sentiment scores to determine the overall sentiment of a piece of text. The model also takes into account the intensity and polarity of words and the presence of negations and other grammatical features to improve the accuracy of sentiment analysis. VADER has been shown to perform well in a variety of applications, including social media analysis, customer feedback analysis, and brand reputation management.

we are implementing the vader model for sentiment analysis. This allows us to evaluate the user's state of mind by analyzing the negativity and positivity of their queries. By using the application over a period of 30 days and implementing the recommended exercises and dietary changes, the user can track changes in their mental health by analyzing the sentiment analysis score on a day-to-day basis.

The proposed application aims to use the Vader model to perform sentiment analysis on user queries, which will help them evaluate their mental state as positive or negative. Users can use the application over a period of 30 days, during which they can implement the recommended exercises and dietary changes to improve their mental health. By analyzing the sentiment analysis scores on a daily basis, users can track their progress and understand the effectiveness of the recommended solutions.

This application will provide users with a valuable tool to monitor their mental health and track their progress over time. It will help users gain a better understanding of their emotional state and provide them with actionable recommendations to improve their mental wellbeing.

The purpose of this application is to provide users with a tool to monitor their mental well-being and make lifestyle changes that can positively impact their mental health. By providing insights into the user's emotional state and offering solutions, the application can help individuals take control of their mental health and improve their overall quality of life.

The VADER (Valence Aware Dictionary and sEntiment Reasoner) model is a lexicon and rule-based sentiment analysis tool that assigns sentiment scores to text based on the presence of certain words and their context.
In VADER, sentiment scores range from -1 to 1, where a score of 1 indicates the most positive sentiment, a score of -1 indicates the most negative sentiment, and a score of 0 indicates a neutral sentiment.

The pos, neg, and neu scores represent the probabilities that the text expresses a positive, negative, or neutral sentiment, respectively. These scores are calculated based on the number of positive, negative, and neutral words in the text, as well as the intensity of those words.

The compound score represents an overall sentiment score calculated based on the other three scores. It ranges from -1 to 1, where a score of -1 indicates the most negative sentiment, a score of 1 indicates the most positive sentiment, and a score of 0 indicates a neutral sentiment. The compound score is calculated using a formula that takes into account the normalized scores of the individual words in the text, as well as their degree of intensity and polarity.

The code for implementing the Gpt model and sentiment analysis model:

```
import time
from django.shortcuts import render
import openai
import nltk
import os

from response.models import Condition
nltk.download('vader_lexicon')
from nltk.sentiment.vader import SentimentIntensityAnalyzer
# Create your views here.
def sentimentAnalysis(explain):
    vader = SentimentIntensityAnalyzer()
    sa=vader.polarity_scores(explain)
    return sa
# Set up OpenAI API key and model
# openai.api_key = "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"
model_engine = "gpt-3.5-turbo-0301"

# openai.api_key = os.getenv("sk-
uo26RWBxAwwsujW3xa4CT3BlbkFJQq3IWJQV8qPXYng2kBzA")
# Define function to send queries to ChatGPT and return responses
def get_gpt_response(request,disorder):
    # disorder = request.POST.get('disorder')
    openai.api_key = "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"
    # disordermod="I am having <symptom> symptoms, can you suggest me the
caution to this, and solutions.".replace("<symptom>",disorder)
    disordermod=disorder
    print(disordermod)
    response = openai.ChatCompletion.create(
      model = model_engine,
    messages=[
        {"role": "user", "content": disordermod}
    ]
    )
    # res = str(response.choices[0].message.content)
    res = response.choices[0].message.content
    res = res[res.index('.')+1:]
    # res = res[res.index(',')+1:]
    # print(response.choices[0].message.content)
    print(res)
    return res


def index(request):
    if request.method == 'POST':
        # print(request.POST)
```

```
        # Get the disorder from the form data
        disorder = request.POST['disorder']
        Symptoms = request.POST.getlist('Symptoms')
        Symptoms = ','.join(Symptoms)
        Gender = request.POST['Gender']
        Age = request.POST['Age']
        explain = f'I am having {disorder} with symptoms {Symptoms}, i am a
{Gender} of age {Age} please try to diagnoise the possible deseases'
        sol = f'I suffering from {disorder} give solution for the given
problem with empathy and accuracy and active listening and  tell me the
possible cures'
        exec = f'I am having {disorder}, suggest me some exercise'
        foodItems = f'I am having {disorder}, suggest me some food Items'
        sentimental = f'I am having {disorder}, give me sentiment analysis'
        # Replace with the actual API endpoint and your API key
        explaination = get_gpt_response(request,explain)
        sa=sentimentAnalysis(explain)
        time.sleep(10)
        solution = get_gpt_response(request,sol)
        time.sleep(10)
        exercise = get_gpt_response(request,exec)
        time.sleep(10)
        food = get_gpt_response(request,foodItems)
        # time.sleep(10)
        # sentiment = get_gpt_response(request,sentimental)
        if disorder:
            # Render the response in a template
            condition = Condition(
                disorder = disorder,
                symptoms = Symptoms,
                age = Age,
                gender = Gender,
                explanation = explaination,
                solution = solution,
                exercise = exercise,
                food = food,
                sentiment = sa
            )
            condition.save()
            sastate=f'The sentiment Anlaysis for the given Query: Positive
score:{sa["pos"]}, Negative Score:{sa["neg"]}, Neutral Score:{sa["neu"]},
Compound Score:{sa["compound"]}'
            return render(request, 'response\\response.html',
{'explain':explain,'disorder':disorder,'Symptoms':Symptoms,'explaination':
explaination,'solution':solution,'exercise':exercise,'food':food,
'sentiment':sastate})
            # return render(request, 'response\\response.html')
        else:
```

```
            # Handle the API error
            return render(request, 'error.html', {'error': 'Error Occurred'})
    else:
        # Render the form for GET requests
        return render(request, 'response\index.html')
```

The explanation and the purpose of the given code is:

The code is a Django-based web application that takes user input about a disorder along with other details such as symptoms, gender, and age. The application uses OpenAI's GPT-3.5 Turbo model to generate responses to user queries related to the given disorder. The responses generated include explanations, solutions, exercises, and food items related to the disorder.

The code defines a function named ‘sentimentAnalysis’ that performs sentiment analysis using the ‘SentimentIntensityAnalyzer’ from the Natural Language Toolkit (NLTK) library. This function takes a text as input and returns a dictionary containing the positive, negative, neutral, and compound scores for the text.
the function sentimentAnalysis is defined, which takes in the parameter explain as the text for which sentiment analysis needs to be performed.

Within this function, an instance of the SentimentIntensityAnalyzer class from the nltk.sentiment.vader module is created. The SentimentIntensityAnalyzer is a pre-trained model that is capable of analyzing the sentiment of a given text.

Next, the polarity_scores() function of the SentimentIntensityAnalyzer class is called on the explain parameter, which returns a dictionary of scores for the given text. The dictionary contains four scores: pos (positive), neg (negative), neu (neutral), and compound.

These scores represent the sentiment of the given text. The ‘positive’ score represents the probability that the text expresses a positive sentiment, the ‘negative’ score represents the probability that the text expresses a negative sentiment, the ‘neutral’ score represents the probability that the text expresses a neutral sentiment, and the ‘compound’ score represents an overall sentiment score calculated based on the other three scores.

Finally, the sentimentAnalysis function returns the dictionary of scores for the given text.

This sentiment analysis is used in the index function to perform sentiment analysis on the explain parameter and display the sentiment scores in the HTML template for the given text.

The index function is the main view that handles GET and POST requests from the user. When a POST request is received, the function gets the disorder, symptoms, gender, and age from the form data submitted by the user. It then generates four different queries using the input data and passes each of them to the GPT-3.5 Turbo model via the ‘get_gpt_response’ function.

The ‘get_gpt_response’ function takes a query and sends it to the ‘OpenAI’ API endpoint to generate a response. It then returns the generated response back to the index function.

The ‘openai.ChatCompletion.create()’ function sends a query to the GPT-3 model via the API and waits for a response. The response is then processed and returned to the user.
The response is then displayed on the frontend of the web application.

The API is essential in this code because it enables the code to leverage the powerful capabilities of the GPT-3 language model. Without the API, the code would not be able to interact with the GPT-3 model and generate responses to the user's queries.

After generating the responses, the index function saves all the details related to the user's input and the generated responses in a database using the Condition model. It then renders the response.html template with the input details and generated responses passed as context variables.

The purpose of the following code is:

```
#!/usr/bin/env python

"""Django's command-line utility for administrative tasks."""
import os
import sys


def main():
    os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'myproject.settings')
    try:
        from django.core.management import execute_from_command_line
    except ImportError as exc:
        raise ImportError(
            "Couldn't import Django. Are you sure it's installed and "
            "available on your PYTHONPATH environment variable? Did you "
            "forget to activate a virtual environment?"
        ) from exc
    execute_from_command_line(sys.argv)


if __name__ == '__main__':
    main()
```

This code is a Python script that serves as a command-line utility for managing Django projects. It is the default script that is generated when you create a new Django project.

The main() function is the entry point of the script, which sets the DJANGO_SETTINGS_MODULE environment variable to the myproject.settings module. This tells Django which settings module to use for the project.

The try block imports the execute_from_command_line() function from the django.core.management module. This function is responsible for parsing command-line arguments and executing the appropriate action in response to the command.

If the execute_from_command_line() function cannot be imported due to an ImportError, the script raises a custom error message that suggests the user check whether Django is installed and whether a virtual environment is activated.

The if *__name__* == '*__main__*' block checks whether the script is being run as the main program, and if so, it calls the main() function, which starts the command-line interface for managing the Django project.

This script is a useful utility for running Django commands from the command line, which is a common way of managing and deploying Django projects.

The purpose of the following code is:

```
<!DOCTYPE html>

<html>
  <head>
    <title>MY PROJECT</title>
    <style>
      /* Style the form container */
.rendered-form {
  max-width: 500px;
  margin: 0 auto;
  padding: 20px;
  background-color: #f2f2f2;
  border-radius: 5px;
  font-family: Arial, sans-serif;
}

/* Style the form headings */
.formbuilder-select-label,
.formbuilder-checkbox-group-label,
.formbuilder-radio-group-label,
.formbuilder-number-label {
  display: block;
  font-weight: bold;
  margin-bottom: 10px;
}

/* Style the form inputs */
.form-group {
  margin-bottom: 20px;
}

/* Style the select input */
.disorder {
  display: block;
  width: 100%;
  padding: 10px;
  font-size: 16px;
  border: none;
  border-radius: 5px;
  background-color: #fff;
  box-shadow: 0 0 5px rgba(0, 0, 0, 0.1);
}

/* Style the checkbox inputs */
.checkbox-group {
  margin-top: 5px;
}
```

```
.formbuilder-checkbox {
  display: inline-block;
  margin-right: 10px;
}

.formbuilder-checkbox input[type="checkbox"] {
  margin-right: 5px;
}

.formbuilder-checkbox label {
  font-size: 14px;
}

/* Style the radio inputs */
.radio-group {
  margin-top: 5px;
}

.formbuilder-radio {
  display: inline-block;
  margin-right: 10px;
}

.formbuilder-radio input[type="radio"] {
  margin-right: 5px;
}

.formbuilder-radio label {
  font-size: 14px;
}

/* Style the number input */
.Age {
  display: block;
  width: 100%;
  padding: 10px;
  font-size: 16px;
  border: none;
  border-radius: 5px;
  background-color: #fff;
  box-shadow: 0 0 5px rgba(0, 0, 0, 0.1);
}

/* Style the submit button */
.submit {
  display: block;
  width: 100%;
  padding: 10px;
```

```
  font-size: 16px;
  border: none;
  border-radius: 5px;
  background-color: #4CAF50;
  color: #fff;
  cursor: pointer;
  transition: background-color 0.3s ease;
}

.submit:hover {
  background-color: #3e8e41;
}

    </style>
    {% comment %} <link rel="stylesheet" href="response\style.css"> {%
endcomment %}
  </head>
  <body style="background-image: url('https://i.ibb.co/Swm68bS/anthony-tran-v-
Xymirxr5ac-unsplash.jpg');background-size: cover;" class="container-fluid">
    {% comment %} <form action="{% url 'index' %}" method="POST">
       {% csrf_token %}
      <label for="query">Enter your query:</label>
      <input type="text" id="query" name="query">
      <button type="submit">Submit</button>
    </form>  {% endcomment %}
    <form method="post">
    <div class="rendered-form">
      {% csrf_token %}
      <div class="formbuilder-select form-group field-disorder">
        <label for="disorder" class="formbuilder-select-label">What disorder
are you suffering from?</label>
        <select class="disorder" name="disorder" id="disorder">
          <option value="Depression" selected="true" id="disorder-
0">Depression</option>
          <option value="Anxiety disorders" id="disorder-1">Anxiety
disorders</option>
          <option value="Bipolar disorder" id="disorder-2">Bipolar
disorder</option>
          <option value="Schizophrenia" id="disorder-3">Schizophrenia</option>
          <option value="Obsessive-compulsive disorder (OCD)" id="disorder-
4">Obsessive-compulsive disorder (OCD)</option>
          <option value="Post-traumatic stress disorder (PTSD)" id="disorder-
5">Post-traumatic stress disorder (PTSD)</option>
          <option value="Eating disorders (anorexia, bulimia, binge-eating
disorder)" id="disorder-6">Eating disorders (anorexia, bulimia, binge-eating
disorder)</option>
          <option value="Attention-deficit/hyperactivity disorder (ADHD)"
id="disorder-7">Attention-deficit/hyperactivity disorder (ADHD)</option>
```

```
          <option value="Borderline personality disorder" id="disorder-
8">Borderline personality disorder</option>
          <option value="Panic disorder" id="disorder-9">Panic
disorder</option>
          <option value="Generalized anxiety disorder (GAD)" id="disorder-
10">Generalized anxiety disorder (GAD)</option>
          <option value="Substance use disorders (alcohol, drugs)"
id="disorder-11">Substance use disorders (alcohol, drugs)</option>
          <option value="Social anxiety disorder (social phobia)"
id="disorder-12">Social anxiety disorder (social phobia)</option>
          <option value="Autism spectrum disorders (ASD)" id="disorder-
13">Autism spectrum disorders (ASD)</option>
          <option value="Insomnia and other sleep disorders." id="disorder-
14">Insomnia and other sleep disorders.</option>
        </select>
      </div>
      <div class="formbuilder-checkbox-group form-group field-Symptoms">
        <label for="Symptoms" class="formbuilder-checkbox-group-label">What
are the Symptoms</label>
        <div class="checkbox-group">
          <div class="formbuilder-checkbox">
            <input class="Symptoms" name="Symptoms[]" access="false"
id="Symptoms-0" value="Depression" type="checkbox" checked="checked">
            <label for="Symptoms-0">Depression</label>
          </div>
          <div class="formbuilder-checkbox">
            <input class="Symptoms" name="Symptoms[]" access="false"
id="Symptoms-1" value="Anxiety disorders" type="checkbox">
            <label for="Symptoms-1">Anxiety disorders</label>
          </div>
        </div>
      </div>
      <div class="formbuilder-radio-group form-group field-Gender">
        <label for="Gender" class="formbuilder-radio-group-
label">Gender</label>
        <div class="radio-group">
          <div class="formbuilder-radio">
            <input class="Gender" name="Gender" access="false" id="Gender-0"
value="Male" type="radio">
            <label for="Gender-0">Male</label>
          </div>
          <div class="formbuilder-radio">
            <input class="Gender" name="Gender" access="false" id="Gender-1"
value="Female" type="radio">
            <label for="Gender-1">Female</label>
          </div>
        </div>
      </div>
```

```
      <div class="formbuilder-number form-group field-Age">
        <label for="Age" class="formbuilder-number-label">Age</label>
        <input type="number" class="Age" name="Age" access="false" min="1"
max="150" id="Age"> </div>
      <div class="formbuilder-button form-group field-submit">
        <button type="submit" class="submit btn-default btn" name="submit"
access="false" style="default" id="submit">Submit</button>
      </div>
    </div>
  </form>
    <script>
      const disorderSelect = document.getElementById('disorder');
      const symptomsCheckboxes = document.querySelector('.checkbox-group');

      // Define the symptoms for each disorder
      const symptomsMap = {
        'Depression': ['Sadness', 'Hopelessness', 'Loss of
interest','Sadness','Hopelessness','Fatigue or lack of energy','Insomnia or
oversleeping','Feelings of worthlessness or guilt','Thoughts of death or
suicide'],
        'Anxiety disorders': ['Excessive worry', 'Fear', 'Panic
attacks','Restlessness','Fatigue','Difficulty
concentrating','Irritability','Muscle tension','Sleep disturbances','Panic
attacks','Avoidance','Nausea','Dizziness','Gastrointestinal problem'],
        'Bipolar disorder': ['Extreme
happiness','Hyperactive','Restlessness','Talking more than usual','Rapid
speech','Racing thoughts','Impulsivity','Feeling sad','Feeling
hopeless','Apathy','Weight changes','Sleep disturbances','Fatigue','Feelings
of guilt','Difficulty focusing','Thoughts of suicide', 'Mood swings'],
        'Schizophrenia': ['Sadness', 'Hopelessness', 'Loss of
interest','Hallucinations or hearing voices','Abnormal motor behavior or
movements', 'Social withdrawal or isolation','Apathy or emotional flatness',
'Paranoia or excessive suspicion'],
        'Obsessive-compulsive disorder (OCD)': ['Fear of contamination or
dirt','Excessive concerns about order, symmetry, or exactness','Excessive
cleaning or hand washing','Repeatedly checking things like locks or
appliances','Repetitive actions like touching or tapping', 'Difficulty
functioning in work, school, or social situations due to symptoms.',
'Distress, anxiety, or guilt related to obsessions or compulsions'],
        'Post-traumatic stress disorder (PTSD)': ['Sadness', 'Hopelessness',
'Loss of interest'],
        'Eating disorders (anorexia, bulimia, binge-eating disorder)':
['Sadness', 'Hopelessness', 'Loss of interest'],
        'Attention-deficit/hyperactivity disorder (ADHD)': ['Sadness',
'Hopelessness', 'Loss of interest'],
        'Borderline personality disorder': ['Sadness', 'Hopelessness', 'Loss
of interest'],
        'Panic disorder': ['Sadness', 'Hopelessness', 'Loss of interest'],
```

```
        'Generalized anxiety disorder (GAD)': ['Sadness', 'Hopelessness',
'Loss of interest'],
        'Substance use disorders (alcohol, drugs)': ['Sadness',
'Hopelessness', 'Loss of interest'],
        'Social anxiety disorder (social phobia)': ['Sadness', 'Hopelessness',
'Loss of interest'],
        'Autism spectrum disorders (ASD)': ['Sadness', 'Hopelessness', 'Loss
of interest'],
        'Insomnia and other sleep disorders.': ['Sadness', 'Hopelessness',
'Loss of interest'],
      };

      // Function to update the checkboxes based on the selected disorder
      function updateSymptomsCheckboxes() {
        const disorder = disorderSelect.value;
        const symptoms = symptomsMap[disorder];

        // Remove any existing checkboxes
        while (symptomsCheckboxes.firstChild) {
          symptomsCheckboxes.removeChild(symptomsCheckboxes.firstChild);
        }

        // Add a checkbox for each symptom
        for (const symptom of symptoms) {
          const checkbox = document.createElement('div');
          checkbox.classList.add('formbuilder-checkbox');
          checkbox.innerHTML = `
            <input class="Symptoms" name="Symptoms" access="false"
value="${symptom}" type="checkbox">
            <label>${symptom}</label>
          `;
          symptomsCheckboxes.appendChild(checkbox);
        }
      }

      // Call the function once to set the initial checkboxes
      updateSymptomsCheckboxes();

      // Call the function whenever the disorder is changed
      disorderSelect.addEventListener('change', updateSymptomsCheckboxes);
    </script>
    {% comment %} <script>
      const disorderSelect = document.getElementById('disorder');
      const symptomsCheckboxes = document.getElementById('symptoms-
checkboxes');

      // Define the symptoms for each disorder
      const symptomsMap = {
```

```
        'Depression': ['Sadness', 'Hopelessness', 'Loss of interest'],
        'Anxiety disorders': ['Excessive worry', 'Fear', 'Panic attacks'],
        'Bipolar disorder': ['Mania', 'Depression', 'Mood swings']
      };

      // Function to update the checkboxes based on the selected disorder
      function updateSymptomsCheckboxes() {
        const disorder = disorderSelect.value;
        const symptoms = symptomsMap[disorder];

        // Remove any existing checkboxes
        while (symptomsCheckboxes.firstChild) {
          symptomsCheckboxes.removeChild(symptomsCheckboxes.firstChild);
        }

        // Add a checkbox for each symptom
        for (const symptom of symptoms) {
          const checkbox = document.createElement('div');
          checkbox.classList.add('formbuilder-checkbox');
          checkbox.innerHTML = `
            <input class="Symptoms" name="Symptoms[]" access="false"
value="${symptom}" type="checkbox">
            <label>${symptom}</label>
          `;
          symptomsCheckboxes.appendChild(checkbox);
        }
      }

      // Call the function once to set the initial checkboxes
      updateSymptomsCheckboxes();

      // Call the function whenever the disorder is changed
      disorderSelect.addEventListener('change', updateSymptomsCheckboxes);
    </script> {% endcomment %}

    <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-
alpha3/dist/js/bootstrap.bundle.min.js" integrity="sha384-
ENjdO4Dr2bkBIFxQpeoTz1HIcje39Wm4jDKdf19U8gI4ddQ3GYNS7NTKfAdVQSZe"
crossorigin="anonymous"></script>
  </body>
</html>
```

This code defines a JavaScript object symptomsMap that contains an array of symptoms for various mental health disorders. Each mental health disorder is represented by a key, and its corresponding value is an array of symptoms associated with that disorder.

The mental health disorders included in this object are: Depression, Anxiety disorders, Bipolar disorder, Schizophrenia, Obsessive-compulsive disorder (OCD), Post-traumatic stress

disorder (PTSD), Eating disorders (anorexia, bulimia, binge-eating disorder), Attention-deficit/hyperactivity disorder (ADHD), Borderline personality disorder, Panic disorder, Generalized anxiety disorder (GAD), Substance use disorders (alcohol, drugs), Social anxiety disorder (social phobia), Autism spectrum disorders (ASD), and Insomnia and other sleep disorders.

The code defines a function updateSymptomsCheckboxes which is called whenever the user changes the value of a select element with the ID 'disorder'. This function updates a list of checkboxes with the symptoms associated with the selected disorder.

The symptoms associated with each disorder are defined in the symptomsMap object. This object has keys corresponding to the names of each disorder, and values that are arrays of symptom strings.

The function updateSymptomsCheckboxes first gets the currently selected disorder from the select element with the ID 'disorder', and then gets the corresponding array of symptoms from symptomsMap. It then removes any existing checkboxes from the list of checkboxes with the ID 'symptoms-checkboxes', and adds a new checkbox for each symptom in the array.

Each checkbox is created as a div element with a class of 'formbuilder-checkbox'. It contains an input element with a type of 'checkbox', a name attribute of 'Symptoms', and a value attribute that is set to the current symptom being iterated over in the loop. It also contains a label element with the text of the current symptom.

Finally, the function is called once when the page loads to set the initial checkboxes based on the initial value of the disorder select element, and then it is also set to be called again whenever the user changes the value of the disorder select element using an event listener that listens for a 'change' event on the select element.

The script begins with a function called updateSymptomsCheckboxes(). This function is called whenever the disorder dropdown menu is changed, and it updates the checkboxes with the symptoms for the selected disorder. It first gets the currently selected disorder and the associated symptoms from a symptomsMap object, which contains an array of symptoms for each disorder. It then removes any existing checkboxes and generates new checkboxes for each symptom using a for loop. Each checkbox is created using document.createElement(), and the classList.add() method is used to add a CSS class to the checkbox. The checkbox is given an input element with type="checkbox", name="Symptoms[]", value="${symptom}", and access="false". It also has a label element that displays the symptom name. Finally, the checkboxes are appended to the symptomsCheckboxes element using appendChild().

The updateSymptomsCheckboxes() function is called once initially to set the checkboxes to the symptoms for the default disorder. The addEventListener() method is used to add an event listener to the disorder dropdown menu that calls updateSymptomsCheckboxes() whenever the selected disorder changes.The document also includes a comment block that begins with {% comment %} and ends with {% endcomment %}.
This is a special syntax used by the Liquid templating language, which is used by some web

platforms like Shopify to generate dynamic content. The content within the comment block is ignored by the browser and does not affect the functionality of the script.

EXPERIMENTAL SETUP

The screenshot of the application:

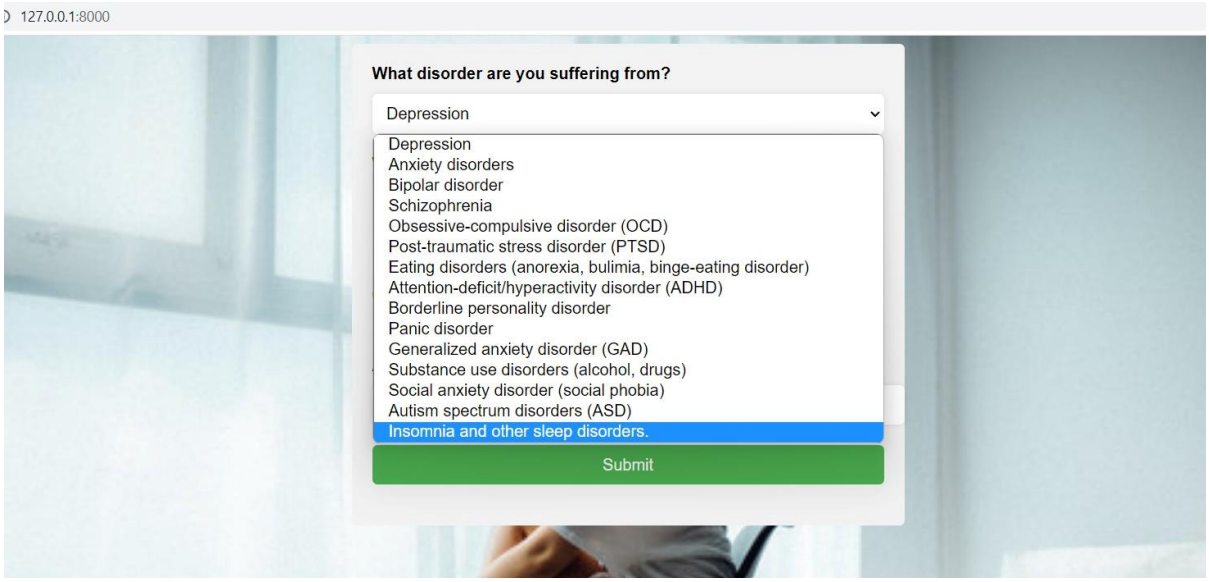


Figure 3.2

The *figure3.2* represents the drop-down of the web application where the user can select the disorder.

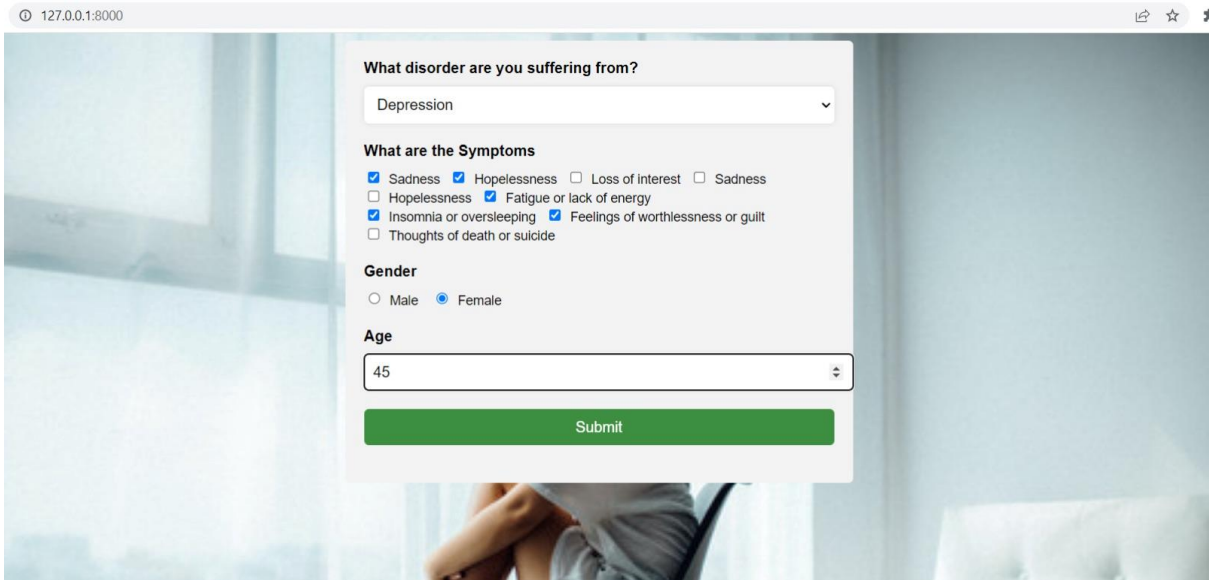


Figure 3.3

The *figure3.3* represents the check box option of the web application where the user can select the symptoms which they are having.

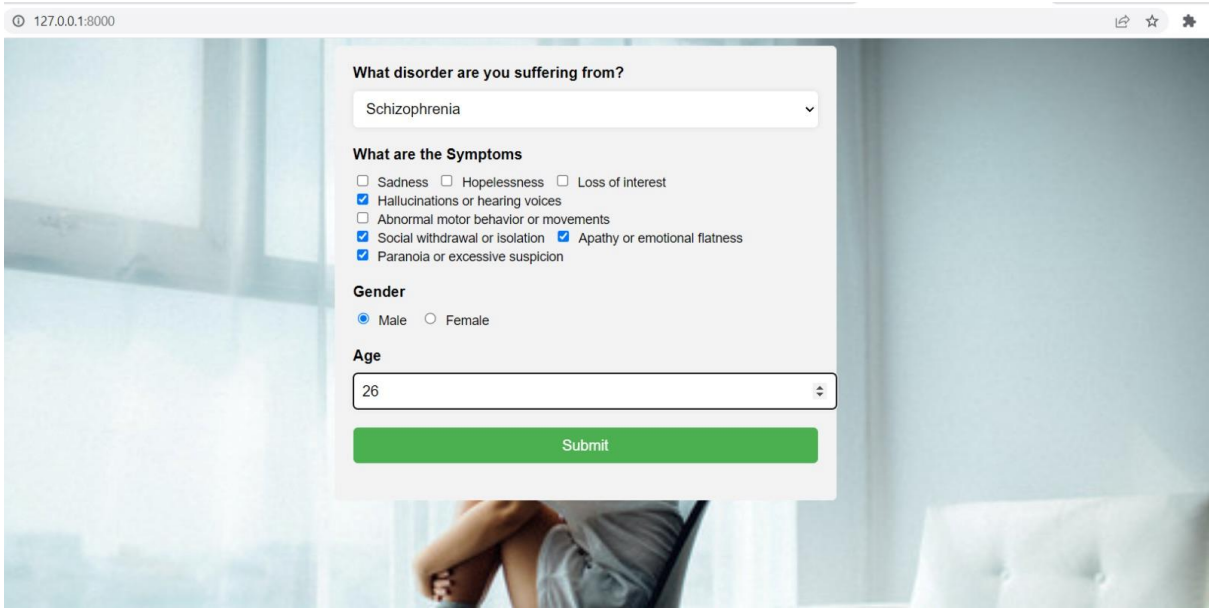


Figure 3.4

The *figure3.4* represents the radio button option of the web application where the user can select their Gender.

Note: The purpose of mentioning the gender is to evaluate whether the application can give more specific solution based on gender.

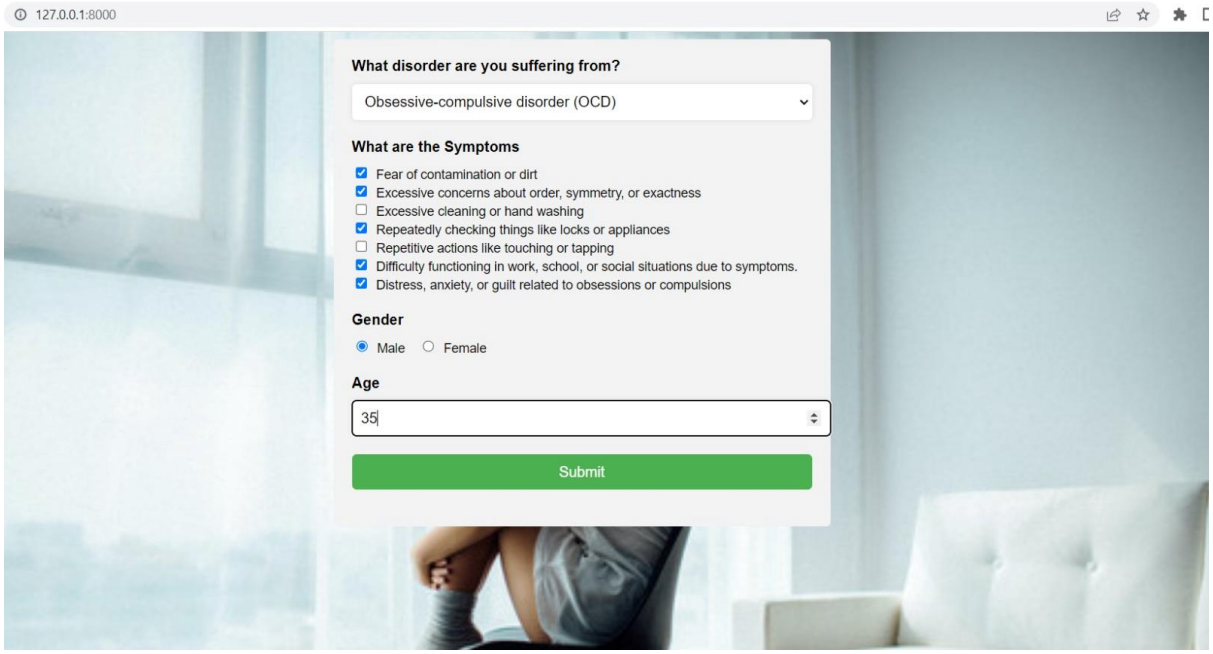


Figure 3.5

The *figure3.5* represents the option to enter the age of the user in the web application.

Note: The purpose of mentioning the Age is to evaluate whether the application can give solution based on the user age, if the user is under age or a child will the model treat the user differently.

The Solution to the User query are as follow:

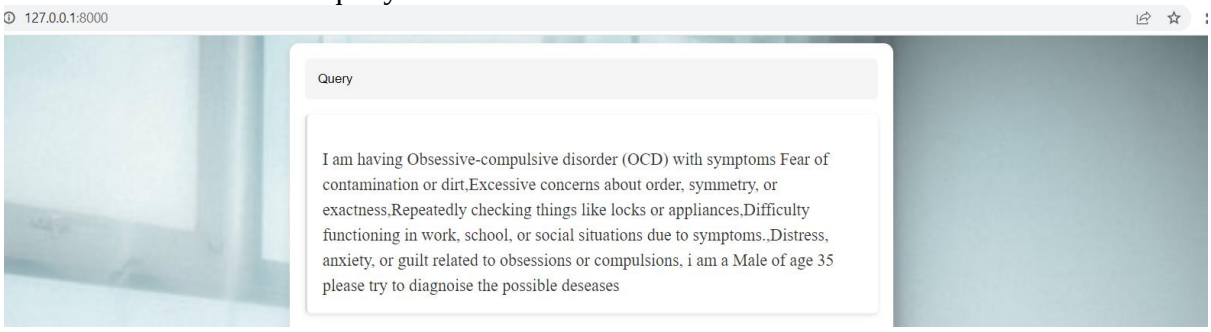


Figure 3.6

The representation of the *figure 3.6* shows the user query.

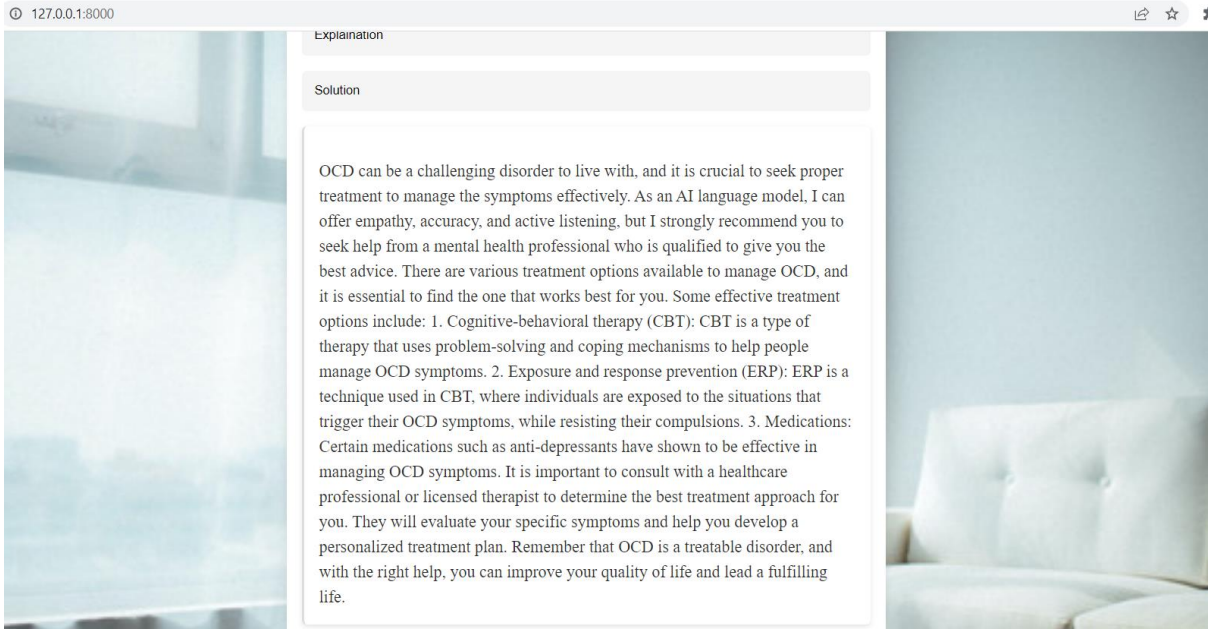


Figure 3.7

The representation of the *figure 3.7* shows the solution given by the web application to the user query.

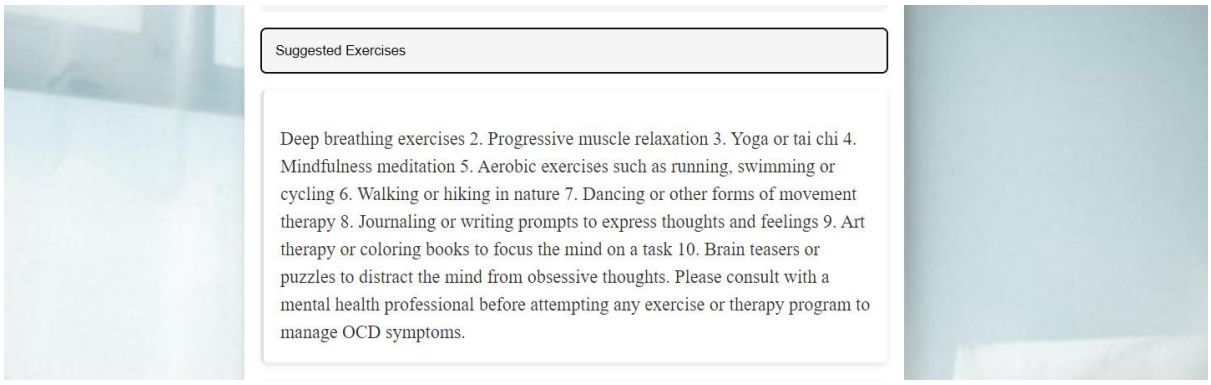


Figure 3.8

The representation of the *figure 3.8* shows the exercise recommendation given by the web application to the user query to improve the physical and the mental health of the user.

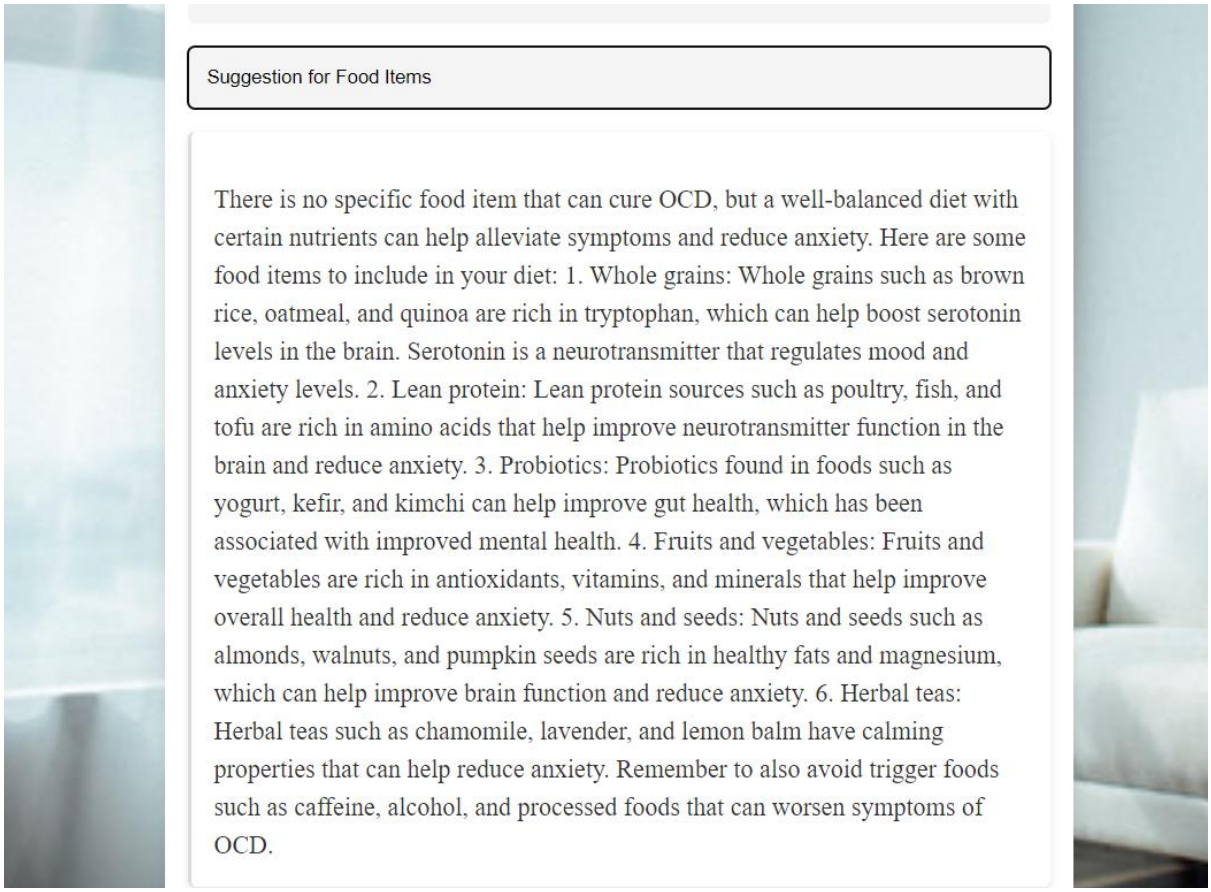


Figure 3.9

The representation of the *figure 3.9* shows the Diet recommendation given by the web application to the user query to improve the physical and the mental health of the user.

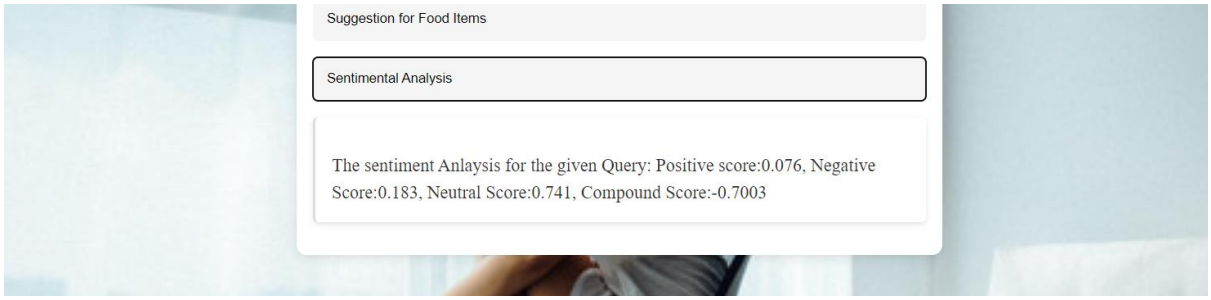
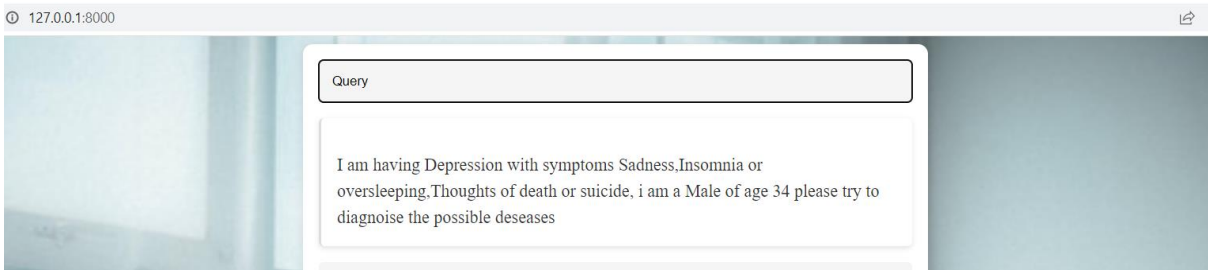


Figure 3.10

The representation of the *figure 3.10* shows the sentiment analysis score of the user query given by the web application to the user to track their mental health status.

Use case 2:



Solution

It is understandable to feel overwhelmed and defeated. You are not alone, and there is help and hope available. Firstly, it is essential to reach out for professional help. You can talk to a doctor or a mental health professional who can provide a diagnosis and offer various treatment options. They may suggest medication, psychotherapy, or a combination of both. In addition to professional help, here are some suggestions that might help ease the symptoms of depression: 1. Reach out to friends and family: Talk to someone who cares about you. Share your feelings and concerns with them. 2. Exercise: Physical activity is an effective way to reduce stress, boost mood, and promote overall health. 3. Practice good sleep habits: Getting enough sleep is crucial for emotional and physical health. Create a comfortable sleep environment and maintain a regular sleep schedule. 4. Engage in pleasurable activities: Participate in activities that you enjoy and find fulfilling, such as reading, watching movies, or pursuing a hobby. 5. Eat a healthy diet: Eating healthy food can help stabilize mood and improve overall health. Remember that depression is a treatable condition, and it is possible to feel better with the right help and support. Do not hesitate to seek the assistance you need to overcome depression.

Solution

Suggested Exercises

Therefore, if you are experiencing depression, it's recommended to consult with a medical professional to get the right treatment plan that suits your condition. However, exercise is a proven method to improve mood and alleviate symptoms of depression. Here are some exercises that you can consider: 1. Aerobic exercise: Walking, jogging, cycling, swimming, and dancing are some examples of aerobic exercises that can improve mood and reduce symptoms of depression. 2. Yoga: Yoga is a gentle form of exercise that can help reduce stress, anxiety, and symptoms of depression. It can also improve flexibility and body strength. 3. Strength training: Lifting weights or doing bodyweight exercises like push-ups, squats, and lunges can improve mood and reduce symptoms of depression. 4. Mindfulness activities: Activities such as mindfulness meditation, tai chi, and qigong can help reduce stress, increase relaxation, and improve mood. Remember, it's essential to find an exercise that you enjoy and can sustain in the long term. Start small and gradually increase the frequency and intensity of your exercise routine. Also, be sure to consult with your medical professional before starting any exercise program.

Suggestion for Food Items

I suggest consulting a doctor or a registered dietician, who can provide you with more customized recommendations based on your specific needs. Additionally, depression is a medical condition that requires professional treatment, so I recommend seeking professional help and support.

Sentimental Analysis

The sentiment Anlaysis for the given Query: Positive score:0.067, Negative Score:0.352, Neutral Score:0.581, Compound Score:-0.8957

Implementing the Comparison Analysis on the web application:

We implemented a Natural Language Toolkit-based paragraph comparison algorithm to compare the answers generated by ChatGPT, human participants, and other chatbots selected from a drop-down option in the web application.

Natural Language Toolkit (NLTK) is a popular Python library used for natural language processing tasks. One of the functionalities of NLTK is to compare two paragraphs of text to determine their similarity or dissimilarity.

The NLTK library provides several methods for comparing two paragraphs. One common method is to use the cosine similarity metric, which measures the cosine of the angle between two vectors in a multi-dimensional space. In the context of text comparison, each paragraph is represented as a vector in a high-dimensional space, with each dimension corresponding to a specific word in the text.

To compare two paragraphs using cosine similarity, the paragraphs are first preprocessed to remove stop words (such as "the" and "and") and to convert all words to their base form (using stemming or lemmatization). Then, the frequency of each word in each paragraph is calculated and used to represent the paragraphs as vectors. The cosine similarity between the two vectors is then calculated to give a measure of their similarity.

NLTK also provides other methods for comparing two paragraphs, such as the Jaccard similarity coefficient and the Dice coefficient. These methods also involve calculating the frequency of words in each paragraph and comparing them, but use different formulas to determine the similarity between the two paragraphs.

The following code explains the implementation of the word comparison between the AI answer and the Human answer.

```
$(document).ready(function () {

  // populate the first dropdown
  var $select1 = $("#select1");
  $select1.append("<option value=''>Select type of question</option>");
  $select1.append("<option value='questions'>Questions</option>");
  $select1.append("<option value='realcase'>Real Case</option>");

  var $select2 = $("#questions");
  var $select3 = $("#answer_providers");

  var $answer1 = $("#chatgpt_answer");
  var $answer2 = $("#human_answer");
  var $answer3 = $("#selected_provider_answer");

  var $accuracy1 = $("#chatgpt_answer_accuracy");
  var $accuracy2 = $("#human_answer_accuracy");
```

```
  var $accuracy3 = $("#selected_provider_answer_accuracy");
  // populate the second dropdown
  $select1.change(function () {
    var value = $(this).val();
    console.log(value)

    $.ajax({
      url: "fetch_questions.php",
      data: { isRealCase: value == "questions" ? 0 : 1, selectedOption: value
},
      type: "GET",
      success: function (data) {
        data = JSON.parse(data);
        $select2.empty();
        $select3.empty();
        $answer1.val('');
        $answer2.val('');
        $answer3.val('');
        $select2.append("<option value=''>Select a question</option>");
        if (Array.isArray(data)) {
          data.forEach(function (value) {
            $select2.append("<option value='" + value.question + "'>" +
value.question + "</option>");
          });
        }
      },
      error: function (jqXHR, textStatus, errorThrown) {
        console.log(textStatus, errorThrown);
      }
    });
  });

  // populate the third dropdown
  $select2.change(function () {
    var question = $(this).val();
    if (question != "") {
      $.ajax({
        url: "fetch_answer_providers.php",
        data: { question: question },
        type: "GET",
        success: function (data) {
          data = JSON.parse(data);
          $select3.empty();
          $answer1.val('');
          $answer2.val('');
          $answer3.val('');
          $select3.append("<option value=''>Select an answer
provider</option>");
```

```
          $.each(data, function (index, value) {
            if (value != 'ChatGPT' && value != 'Human')
              $select3.append("<option value='" + value + "'>" + value +
"</option>");
          });
        },
        error: function (jqXHR, textStatus, errorThrown) {
          console.log(textStatus, errorThrown);
        }
      });
    }
  });

  function getMatchingPercentage(paragraph1, paragraph2) {
    // Convert paragraphs to lowercase and split into words
    const words1 = paragraph1.toLowerCase().split(' ');
    const words2 = paragraph2.toLowerCase().split(' ');

    // Get the number of matching words
    const matchingWords = words1.filter(word => words2.includes(word)).length;

    // Calculate the percentage of matching words
    const percentage = Math.round((matchingWords / words1.length) * 100);

    return percentage;
  }


  // display the answers
  $select3.change(function () {
    var questionId = $select2.val();
    var answerProvider = $(this).val();
    if (questionId != "" && answerProvider != "") {
      $.ajax({
        url: "fetch_answers.php",
        data: { questionId: questionId, answerProvider: answerProvider },
        type: "GET",
        success: function (data) {
          data = JSON.parse(data);
          $answer1.val(data.chatgptAnswer);
          $answer2.val(data.humanAnswer);
          $answer3.val(data.selectedAnswerProviderAnswer);

          var chatgptAccuracy =
getMatchingPercentage(data.chatgptAnswer,data.humanAnswer)
          var selectedAnswerProviderAccuracy =
getMatchingPercentage(data.selectedAnswerProviderAnswer,data.humanAnswer)
```

```
          $accuracy1.val(`Accuracy of ChatGPT's answer as compared to Human's
Answer is ${chatgptAccuracy}%`)
          $accuracy2.val("Accuracy of Human's Answer as compared to Human's
Answer is 100%")
          $accuracy3.val(`Accuracy of ${answerProvider}'s answer as compared
to Human's Answer is ${selectedAnswerProviderAccuracy}%`)
        },
        error: function (jqXHR, textStatus, errorThrown) {
          console.log(textStatus, errorThrown);
        }
      });
    }
  });
});
```

This code block is defining a jQuery function that runs when the document is loaded and ready. Its purpose is to populate the first dropdown menu with two options - "Questions" and "Real Case" - and to initialize several variables that will be used later in the code.

The function then defines two more dropdown menus, $select2 and $select3, which are initially empty but will be populated later depending on the selection made in $select1.

The function defines several variables, $answer1, $answer2, $answer3, $accuracy1, $accuracy2, and $accuracy3, which correspond to various elements in the HTML document. These variables will be used later in the code to display information and results.

The purpose of this code in our project is to set up the initial state of the dropdown menus and to define the necessary variables for later use. It also sets up an event listener that detects when the user selects an option from $select1, which will trigger further actions to populate the other dropdown menus and display relevant information.

The $.ajax function is used to make the request, with the url property specifying the URL of the server-side script (fetch_questions.php), and the data property specifying the parameters to send along with the request. The type property specifies the HTTP method to use, which is GET in this case.

The success property specifies a callback function to be called when the request is successful, with the response data passed as a parameter. In this case, the response data is parsed as a JSON object and used to populate the second dropdown list with the fetched questions.

If an error occurs during the request, the error property specifies a callback function to be called, with the error details passed as parameters.
The purpose of this code in our project is to fetch questions from the server based on the selected option in the first dropdown and populate the second dropdown list with the fetched questions.

The code populates the third dropdown based on the selection made in the second dropdown. When the user selects a question from the second dropdown, an AJAX request is made to fetch the answer providers available for that question. The response is then used to populate the third dropdown with the available answer providers. The code also resets the answers and their corresponding accuracy values when the selection in the second dropdown is changed.

The code defines a function called getMatchingPercentage that takes two paragraphs as input and returns the percentage of matching words between the two paragraphs. The function first converts the paragraphs to lowercase and splits them into words. It then counts the number of words that are present in both paragraphs and calculates the percentage of matching words. This function can be used to determine the similarity between two paragraphs of text.

Based on our project, it seems that this code is used to populate dropdown menus and calculate the similarity between paragraphs. The dropdowns likely contain questions and answer providers, and the code is used to dynamically fetch and populate them based on user selection. The getMatchingPercentage function could be used to compare the accuracy of answers provided by different answer providers.

This code handles the event when the third dropdown menu is changed, and it retrieves the answers for the selected question and answer provider using an AJAX request to "fetch_answers.php". It then updates the values of the three answer input fields with the retrieved answers, and calculates the accuracy of each answer using the "getMatchingPercentage" function. Finally, it updates the values of the three accuracy input fields with the calculated accuracies.

In our project, this code is part of a system for retrieving and comparing answers from different sources (such as ChatGPT and human experts) for a given question, and displaying the results to the user. The purpose of this specific code is to handle the selection of an answer provider from the third dropdown menu and to update the answer and accuracy input fields accordingly.

The following code explains the implementation of the word comparison on php using Xampp:

The following code explains how the web application is fetch questions and real cases from a MySQL database

```php
<?php
// Fetch questions from database based on the selected option
$selectedOption = $_REQUEST['selectedOption'];

// Connect to database
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "compare";

$conn = new mysqli($servername, $username, $password, $dbname);

// Check connection
if ($conn->connect_error) {
  die("Connection failed: " . $conn->connect_error);
}

// Prepare SQL query based on selected option
if ($selectedOption === 'questions') {
  $sql = "SELECT DISTINCT question FROM questions WHERE isRealCase = 0";
} else if ($selectedOption === 'realcase') {
  $sql = "SELECT DISTINCT question FROM questions WHERE isRealCase = 1";
} else {
  echo "Invalid option selected.";
  exit();
}

// Execute SQL query
$result = $conn->query($sql);

// Check for errors
if (!$result) {
  echo "Error: " . $sql . "<br>" . $conn->error;
  exit();
}

// Fetch result data as array
$data = array();
while ($row = $result->fetch_assoc()) {
  $data[] = $row;
}
```

```
// Debugging output
// var_dump($data);
// echo "<script> console.log($data);</script>";

// Return data as JSON
echo json_encode($data);

// Close database connection
$conn->close();
?>
```

This PHP code fetches questions from a MySQL database based on the selected option received via the "selectedOption" parameter in the HTTP request. The code connects to the database using the given server name, username, password, and database name. It then prepares an SQL query based on the selected option and executes it to retrieve the question data.

If the query is successful, the data is fetched as an associative array and stored in a variable named $data. The code then encodes this array into JSON format and sends it back as the response to the HTTP request.

This code is an AJAX request handler and is called by the JavaScript code in the project to fetch questions from the server based on the user's selection.
This code is a PHP script that fetches questions from a MySQL database based on the selected option sent as a parameter in the AJAX call. The selected option is obtained using $_REQUEST['selectedOption'].

The code first establishes a connection to the MySQL database using mysqli. It then prepares an SQL query based on the selected option, either to fetch distinct questions that are not real case scenarios, or those that are real case scenarios. The SQL query is executed using $conn->query($sql).

If there are any errors, the script will output an error message with the SQL query and the database error. If the query is successful, the result data is fetched as an associative array using $result->fetch_assoc() and stored in the $data array. Finally, the data is encoded as JSON using json_encode($data) and returned to the AJAX call.

The purpose of this code in the project is to fetch the questions from the database based on the selected option (whether to fetch questions that are real case scenarios or not) and return them to the front-end to populate the first dropdown menu in the compare answers page.

The following code explains how the web application is answers for the questions and real cases from a MySQL database:

```
<?php
// Check if the request method is GET
if ($_SERVER['REQUEST_METHOD'] == 'GET') {

  // Get the selected question, answer provider and answer provider's name
from the request
  $selectedQuestion = $_REQUEST['questionId'];
  $selectedAnswerProvider = $_REQUEST['answerProvider'];

  // Establish a connection to the MySQL database
  $conn = new mysqli('localhost', 'root', '', 'compare');

  // Check if the connection was successful
  if ($conn->connect_error) {
    die('Connection failed: ' . $conn->connect_error);
  }

  // Set the default values for the answers
  $chatgptAnswer = '';
  $humanAnswer = '';
  $selectedAnswerProviderAnswer = '';

  // Fetch the answer where answerProvider is ChatGPT
  $sql = "SELECT answer FROM questions WHERE question = '$selectedQuestion'
AND answerProvider = 'ChatGPT'";
  $result = $conn->query($sql);

  // Check if there is a result and store the answer in $chatgptAnswer
  if ($result->num_rows > 0) {
    $row = $result->fetch_assoc();
    $chatgptAnswer = $row['answer'];
  }

  // Fetch the answer where answerProvider is human
  $sql = "SELECT answer FROM questions WHERE question = '$selectedQuestion'
AND answerProvider = 'human'";
  $result = $conn->query($sql);

  // Check if there is a result and store the answer in $humanAnswer
  if ($result->num_rows > 0) {
    $row = $result->fetch_assoc();
    $humanAnswer = $row['answer'];
  }
```

```
  // Fetch the answer where answerProvider is the selected answer provider
  $sql = "SELECT answer FROM questions WHERE question = '$selectedQuestion'
AND answerProvider = '$selectedAnswerProvider'";
  $result = $conn->query($sql);

  // Check if there is a result and store the answer in
$selectedAnswerProviderAnswer
  if ($result->num_rows > 0) {
    $row = $result->fetch_assoc();
    $selectedAnswerProviderAnswer = $row['answer'];
  }

  // Close the database connection
  $conn->close();

  // Create an array with the answers and encode it to JSON format
  $answers = array(
    'chatgptAnswer' => $chatgptAnswer,
    'humanAnswer' => $humanAnswer,
    'selectedAnswerProviderAnswer' => $selectedAnswerProviderAnswer,
    'selectedAnswerProviderName' => $selectedAnswerProvider
  );
  echo json_encode($answers);
}

?>
```

This code is a PHP script that fetches the answers for a selected question from a MySQL database and returns them as a JSON object. The code checks if the request method is GET and then retrieves the selected question and answer provider from the request. It then establishes a connection to the MySQL database and fetches the answer for the selected question from three different answer providers: ChatGPT, human, and the selected answer provider.

The fetched answers are stored in variables, and a JSON object is created with these variables and returned as the response to the GET request. The JSON object contains the answers from the three answer providers, as well as the name of the selected answer provider. Finally, the database connection is closed.

The purpose of this code in the project is to provide a way for the user to compare the answers from different answer providers for a selected question. The code fetches the answers from the database and returns them in a JSON object, which can be displayed on the webpage.
This code handles a GET request and fetches answers for a specific question from a MySQL database based on the selected answer provider.

The code first checks if the request method is GET and retrieves the selected question and answer provider from the request parameters.

It then establishes a connection to the MySQL database and sets default values for the answers.

The code then fetches the answer for the selected question from the database for three different answer providers - ChatGPT, human, and the selected answer provider. For each answer provider, it runs a SELECT query on the "questions" table with the question and answer provider as the search criteria. If there is a result, the code stores the answer in the corresponding variable.

Once all the answers have been retrieved, the code creates an array containing the answers, the selected answer provider's name, and encodes it to JSON format before outputting it.

Finally, the database connection is closed.Overall, the purpose of this code is to retrieve and return answers for a specific question from the database for different answer providers based on the user's selection.

The following code explains how the web application code is retrieving the available answer providers for a selected question from a MySQL

```php
<?php

// Retrieve the selected question id from the GET request
if (isset($_GET['question'])) {
  $question = $_GET['question'];
} else {
  // If question id is not provided in the GET request, return an error
message
  die("Error: Question ID not provided");
}

// Connect to the database
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "compare";

$conn = new mysqli($servername, $username, $password, $dbname);

// Check connection
if ($conn->connect_error) {
  die("Connection failed: " . $conn->connect_error);
}

// Prepare a SQL query to fetch answer providers for the selected question
$sql = "SELECT DISTINCT answerProvider FROM questions WHERE question =
'$question' ORDER BY answerProvider ASC";
$stmt = $conn->prepare($sql);
$stmt->execute();

// Get the result set from the query
$result = $stmt->get_result();

// Create an array to hold the answer providers
$answer_providers = array();

// Loop through the result set and add each answer provider to the array
while ($row = $result->fetch_assoc()) {
  $answer_providers[] = $row['answerProvider'];
}

// Close the database connection
$conn->close();
```

```
// Return the answer providers as a JSON object
echo json_encode($answer_providers);
?>
```

This code retrieves the available answer providers for a selected question from a MySQL database and returns them as a JSON object.

The code first checks if a question ID is provided in the GET request, and if not, it returns an error message. Then, it establishes a connection to the database and prepares a SQL query to fetch the distinct answer providers for the selected question. The query is executed and the result set is retrieved. The answer providers are stored in an array by looping through the result set, and then the database connection is closed. Finally, the array of answer providers is returned as a JSON object.

The purpose of this code in your project could be to provide a list of available answer providers for a selected question in a user interface, so that the user can choose which answer provider's response to display.

This PHP code performs the following functions:

It checks if a question ID is provided in the GET request using the isset() function. If the ID is provided, it assigns the ID to the $question variable. Otherwise, it prints an error message and stops the execution of the script using the die() function.

It connects to a MySQL database using the mysqli class and the new operator.

It prepares a SQL query to retrieve all the distinct answer providers that have provided an answer for the selected question ID. It orders the results by the answer provider name in ascending order.

It executes the SQL query using the execute() method of the mysqli_stmt class and retrieves the result set using the get_result() method.

It loops through the result set using a while loop and adds each answer provider to an array named $answer_providers.

It closes the database connection using the close() method of the mysqli class.

Finally, it returns the $answer_providers array as a JSON object using the json_encode() function.The purpose of this code is to retrieve all the distinct answer providers for a selected question ID from a MySQL database and return the answer providers as a JSON object.

The screenshot of the application storage of data/Answers:

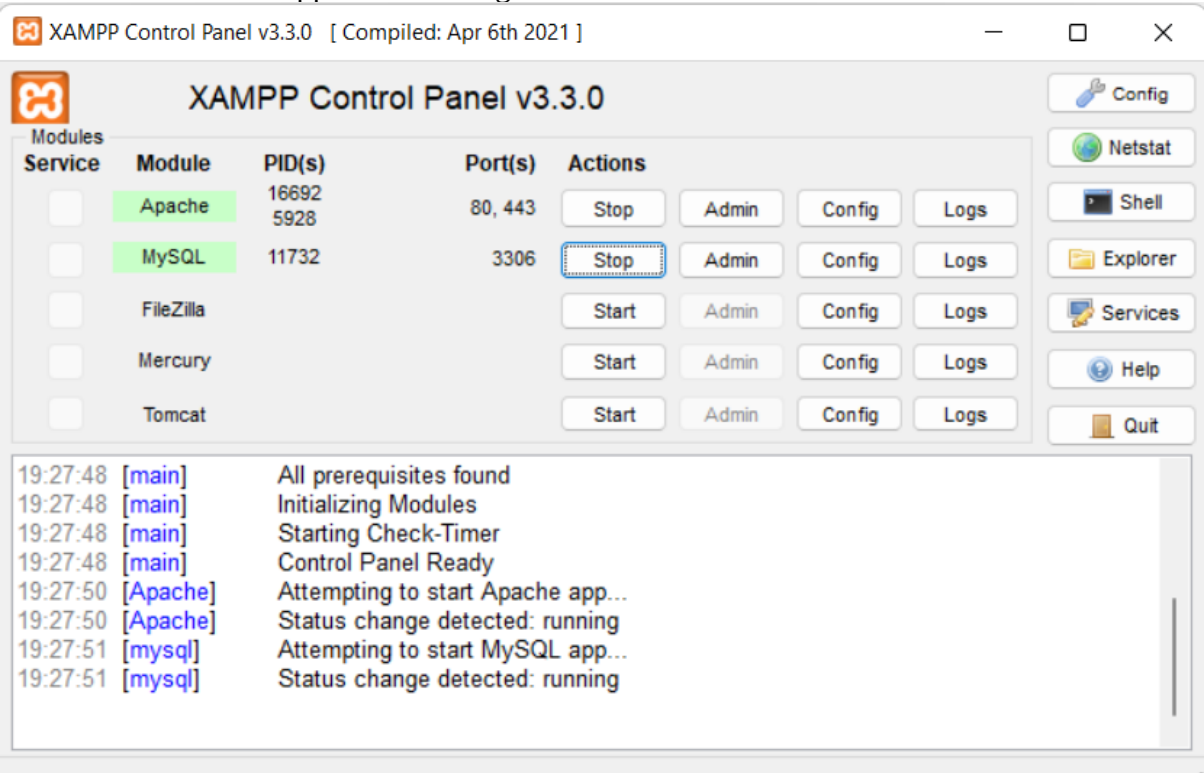


Figure 3.11

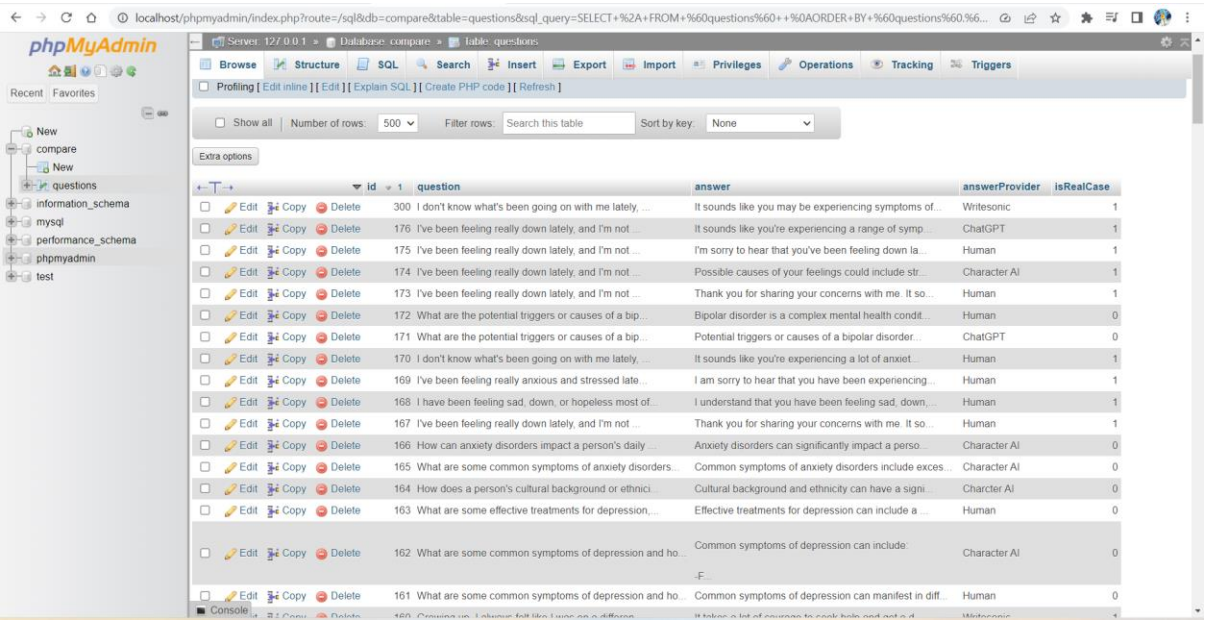


Figure 3.12

The *Figure 3.12* represents the data storage on localhost/phpmyadmin

Result and Analysis

The following are the example of the answer that we collected for the questions and real cases from the AI and Human psychologist:

The Answer from Human and AI to the question:

Question 1: What are some common symptoms of depression and how do they typically manifest in an individual?
Human: symptoms of depression can manifest in different ways for each individual. Generally speaking, some of the most common symptoms of depression include feelings of sadness and/or emptiness, difficulty with concentration, loss of interest in activities that were once enjoyed, changes in appetite or weight, fatigue, difficulty sleeping or sleeping too much, and thoughts of suicide or self-harm. Other physical symptoms such as headaches, stomachaches, and body aches may also be present. Additionally, individuals may experience feelings of low self-worth, guilt, or hopelessness. When interacting with an individual who is exhibiting signs of depression, it is important to provide a supportive and understanding environment. Be sure to actively listen to the individual and reflect back on what they are saying. Show compassion and connect with the individual by making eye contact and being mindful of body language. Provide resources, reassurance, and encouragement. It is also important to be aware of any safety concerns and to provide emotional support.

ChatGPT: Common symptoms of depression may include: Persistent sadness or a low mood Loss of interest or pleasure in activities that were once enjoyable Changes in appetite or weight (either significant weight gain or loss) Insomnia or hypersomnia (difficulty sleeping or sleeping excessively) Fatigue or loss of energy Feelings of worthlessness or excessive guilt Difficulty concentrating or making decisions Thoughts of death or suicide These symptoms may manifest differently in each individual. For example, some people with depression may experience difficulty concentrating or making decisions, while others may have a hard time getting out of bed or performing daily tasks. It's important to note that not everyone with depression will experience all of these symptoms, and that the severity of symptoms can vary from person to person. If you or someone you know is experiencing symptoms of depression, it's important to seek professional help.

AI answer: Youchat: Common symptoms of depression include: Feelings of sadness, hopelessness, emptiness or guilt Loss of interest or pleasure in activities that were once enjoyable Changes in appetite or weight, including overeating or undereating Difficulty concentrating, making decisions or remembering things Sleeping too much or too little Feeling tired or lack of energy Physical symptoms such as headaches, stomachaches or other aches and pains Thoughts of self-harm or suicide Irritability or restlessness Social withdrawal or isolation Depression symptoms can vary from person to person and may present differently based on factors such as age, gender, and cultural background. It's important to seek professional help if you or someone you know is experiencing symptoms of depression.

AI answer: Writesonic: Depression can manifest in a variety of ways, but common symptoms include feelings of sadness, hopelessness, or emptiness, loss of interest in activities once enjoyed, changes in appetite and sleeping patterns, fatigue, difficulty concentrating, and thoughts of self-harm or suicide. Physical symptoms such as headaches, stomachaches, and muscle pain may also be present. Each individual may experience depression differently, but it's important to seek help if symptoms persist for more than two weeks or interfere with daily functioning. Treatment options can help alleviate symptoms and improve quality of life.differently based on factors such as age, gender, and cultural background. It's important to seek professional help if you or someone

you know is experiencing symptoms of depression.

Question 2: What are some effective treatments for depression, and how can individuals with depression seek help or support?

Human: Effective treatments for depression can include a combination of psychotherapy, medication, and lifestyle changes. Psychotherapy, or talk therapy, can help individuals with depression identify, understand, and work through the underlying causes of their depression. Cognitive Behavioral Therapy (CBT), Interpersonal Therapy, and Dialectical Behavioral Therapy are all effective forms of psychotherapy for depression. Medication, such as antidepressants and mood stabilizers, can be used to reduce symptoms of depression. It is important to note, however, that medication alone is not enough to treat depression. It must be used in conjunction with psychotherapy and lifestyle changes in order to be most effective. Lifestyle changes can also be a powerful tool in managing depression. Regular exercise, a healthy diet, adequate sleep, and social support can all help to reduce symptoms of depression. Additionally, engaging in activities that bring joy and meaning to life, such as hobbies and meaningful relationships, can be beneficial in managing depression. Individuals with depression can seek help or support from a variety of sources. A mental health professional, such as a therapist or psychiatrist, can provide individualized treatment and support. Support groups, either in-person or online, can also provide a valuable source of connection and support.

ChatGPT: There are several effective treatments for depression, including: Psychotherapy: Also known as "talk therapy," this involves working with a mental health professional to identify and address the underlying causes of depression. Cognitive-behavioral therapy (CBT) is a specific type of psychotherapy that has been shown to be particularly effective for treating depression. Medication: Antidepressant medications can help alleviate symptoms of depression by rebalancing neurotransmitters in the brain. However, medication is not always necessary and should be used under the guidance of a mental health professional. Lifestyle changes: Engaging in regular exercise, getting enough sleep, eating a healthy diet, and avoiding drugs and alcohol can all help alleviate symptoms of depression. Support groups: Connecting with others who are going through similar experiences can provide a sense of community and reduce feelings of isolation. Individuals with depression can seek help or support by: Talking to a mental health professional, such as a therapist or psychiatrist, who can help develop a treatment plan tailored to their specific needs. Speaking with a primary care physician or other medical professional, who can provide referrals to mental health specialists or prescribe medication if necessary. Reaching out to a crisis hotline, such as the National Suicide Prevention Lifeline, for immediate support and guidance. Connecting with local support groups or online communities for individuals with depression.

AI answer: Character AI: 1. Psychotherapy: Talk therapy is one of the most effective treatments for depression. It can help individuals identify and understand patterns of thinking and behavior that contribute to feelings of depression, and help them to develop healthier ways of coping. 2. Antidepressant medication: Antidepressants work by increasing the amount of certain chemicals in the brain that can help regulate mood. They can be effective in relieving symptoms of depression, but it may take several weeks for them to take full effect. 3. Cognitive-behavioral therapy (CBT): This type of therapy helps individuals to recognize unhelpful patterns of thought and behavior and replace them with more positive and adaptive ones. It can help individuals to cope with difficult emotions and manage symptoms of depression. Individuals with depression can seek help and support by speaking

with their primary care provider or a mental health professional. They can also call a hotline or seek out local support groups.

AI answer: Youchat: Effective treatments for depression include therapy, medication, lifestyle changes, and support from family and friends. Here are some options: Therapy: Psychotherapy, also called talk therapy or counseling, can help individuals with depression by providing them with tools to manage symptoms and cope with stress. Cognitive-behavioral therapy (CBT) is one type of therapy that can be particularly helpful for people with depression. Medication: Antidepressant medications can help relieve symptoms of depression. There are several different types of antidepressants, and finding the right medication can take some trial and error, so it's important to work with a healthcare provider. Lifestyle changes: Exercise, healthy eating habits, regular sleep patterns, and stress-reducing activities such as yoga can all help alleviate symptoms of depression. Support: Support from family, friends, or support groups can aid in managing depression symptoms and provide a sense of community. Individuals with depression can seek help by talking to their healthcare provider or a mental health professional, such as a therapist or counselor. They can also contact organizations such as the National Alliance on Mental Illness (NAMI) or the Depression and Bipolar Support Alliance (DBSA) for resources and support.

AI answer: Writesonic: Depression can be treated effectively, and there are different types of treatments available. Medication is one of the most common treatments, and there are different types of antidepressants available, such as SSRIs, SNRIs, and tricyclic antidepressants. Therapy can also be effective, including talk therapy, cognitive therapy, behavioral therapy, and CBT [2]. Additionally, lifestyle changes such as regular exercise, a healthy diet, and getting enough sleep can also be helpful [3]. If you are experiencing symptoms of depression, it is important to seek help. You can talk to a mental health professional or a healthcare provider. Additionally, there are suicide prevention hotlines such as Mitram Foundation that offer emotional support to those in crisis [1]. Remember, depression is a treatable condition, and seeking help is the first step towards recovery.**Question 3:**

Question 3: How can anxiety disorders impact a person's daily life, and what are some coping strategies that can help manage anxiety symptoms?

Human: Anxiety disorders can have a significant impact on a person's daily life. People with anxiety may experience difficulty sleeping, difficulty concentrating, difficulty with daily tasks, difficulty with social interactions, difficulty with physical activities, and difficulty with decision-making. Coping strategies that can help manage anxiety symptoms include relaxation techniques such as mindfulness, deep breathing, and progressive muscle relaxation; cognitive restructuring to help identify and challenge unhelpful thoughts; regular exercise; healthy nutrition; getting adequate sleep; and engaging in activities that bring pleasure and enjoyment. Additionally, it can be helpful to connect with a mental health professional to develop an individualized plan that can help address anxiety symptoms.

ChatGPT: Anxiety disorders can significantly impact a person's daily life, leading to difficulties in relationships, work or school performance, and other areas of functioning. Some ways that anxiety disorders can impact daily life include: Avoidance behaviors: People with anxiety disorders may avoid certain situations or activities that trigger their anxiety, leading to a restricted and limited life. Interpersonal difficulties: Anxiety can cause irritability, restlessness, and difficulty concentrating, which can negatively impact relationships with family, friends, and colleagues. Physical health problems: Chronic anxiety can lead to physical health problems such as high blood pressure, gastrointestinal problems, and chronic pain. Sleep disturbances: Anxiety can cause difficulty falling asleep, staying asleep, or waking up feeling rested, which can impact overall well-being. There are many coping strategies that can help manage anxiety symptoms, including: Deep breathing and

relaxation techniques, such as meditation or yoga, can help calm the mind and reduce physical symptoms of anxiety. Regular exercise can help reduce symptoms of anxiety and improve overall mood and well-being. Cognitive-behavioral therapy (CBT) is a type of therapy that can help people identify and challenge negative thought patterns that contribute to anxiety. Medication, such as antidepressants or anti-anxiety medications, may be prescribed by a healthcare provider to help manage symptoms. Seeking social support from friends, family, or support groups can help individuals feel less isolated and provide a sense of community. It is important to note that coping strategies for anxiety can vary depending on the individual and the specific type of anxiety disorder they are experiencing. Seeking support from a mental health professional can help individuals develop a personalized treatment plan that addresses their specific needs.

AI answer: Character AI: Anxiety disorders can significantly impact a person's daily life by causing physical and psychological symptoms such as fatigue, insomnia, restlessness, irritability, difficulty concentrating, muscle tension, and headaches. Anxiety can also lead to avoidance behaviors, social isolation, and difficulty completing tasks. Coping strategies that can help manage anxiety symptoms include: • Exercise and physical activity: Exercise can help release endorphins and reduce stress. • Cognitive Behavioral Therapy: This type of talk therapy can help identify and modify maladaptive thoughts and behaviors. • Relaxation techniques such as breathing exercises, progressive muscle relaxation, guided imagery, and yoga. • Cognitive restructuring: This technique helps to reframe negative thoughts and beliefs into more positive and realistic ones. • Meditation: This practice can help to reduce stress and improve focus. • Social support: Talking to family and friends can help mitigate anxiety symptoms. • Journaling: Writing down thoughts and feelings can help to process emotions and provide clarity.

AI answer: Youchat: Anxiety disorders can impact a person's daily life by causing excessive worry, fear, and physical symptoms such as muscle tension and restlessness. This can make it difficult to concentrate, perform daily tasks, and engage in social situations. Coping strategies that can help manage anxiety symptoms include: Relaxation techniques: Deep breathing, meditation, and progressive muscle relaxation can help reduce physical symptoms of anxiety. Exercise: Regular physical activity can help reduce stress and improve mood. Therapy: Cognitive-behavioral therapy (CBT) is particularly helpful for treating anxiety disorders by providing tools to manage symptoms and change negative thought patterns. Medication: Antidepressant medications, particularly selective serotonin reuptake inhibitors (SSRIs), can be effective in treating certain anxiety disorders. Lifestyle changes: Reducing caffeine and alcohol intake, getting enough sleep, and eating a healthy diet can help manage anxiety symptoms. It's important to seek help if anxiety symptoms interfere with daily life, and a mental health professional or healthcare provider can help provide support and guidance on managing anxiety.

AI answer: Writesonic: Anxiety disorders can have a significant impact on a person's daily life, causing excessive worry, fear, and panic in everyday situations. Symptoms may include restlessness, irritability, difficulty concentrating, muscle tension, and sleep disturbances. Good lifestyle practices, such as physical activity, healthy eating, prioritizing sleep, and stress management, can help manage anxiety symptoms. Relaxation techniques, mindfulness, and meditation can also be effective coping strategies for managing anxiety. Additionally, seeking help from a healthcare provider or mental health professional can provide support in identifying triggers and developing healthy coping strategies.

The Answer from Human and AI to the Real Cases :

Real case 1: I've been feeling really down lately, and I'm not sure why. I've been having trouble sleeping and have lost my appetite. I just feel really unmotivated and don't enjoy things the way I used to. It's been going on for several months now. I just can't seem to shake this feeling everything has been pretty stable in my life.

Human: Thank you for sharing your concerns with me. It sounds like you have been feeling down for a while now and it is understandable that this is concerning for you. It is important to talk about how we are feeling and I am glad that you are reaching out for help. There can be many factors that contribute to feeling down and unmotivated. It is important to get a better understanding of what is behind these feelings in order to find the best way to address them. I would suggest that we start by exploring what has been going on for you recently and what may have been causing this change in your mood. It could also be helpful to look at how you typically manage stress, situations in your life that might be contributing to these feelings, and the thoughts and beliefs that you have about yourself and the world around you. It is also important to make sure that you are taking care of your physical health. This can include getting enough rest and nourishing your body with healthy foods. Exercise, yoga, and mindfulness activities can also be helpful in managing mood. I understand that it can be difficult to reach out for help when you feel down. I am here to support you and help you find the best way to manage your feelings. It sounds like you're going through a difficult time right now. It's understandable that you would be feeling down and unmotivated when these changes have been going on for a while. It's important to recognize that it's okay to not feel okay, and that it's normal to have days where you don't feel your best. It's important to take care of yourself in times like these. It may be helpful to focus on things like exercising, eating healthy foods, getting enough rest, and participating in activities that bring you joy. It's also important to talk to someone about how you're feeling. You can talk to a close friend or family member, or you can even speak with a mental health professional if you feel like you need that extra support. No matter what you decide to do, remember that you don't have to go through this alone. There are people who care about you and want to help.

ChatGPT: Possible causes of your feelings could include stress, depression or anxiety. It might be helpful to talk to a mental health professional to figure out what's going on and to get the help you need. It could also be beneficial to practice self-care activities like yoga, meditation, or journaling to help relax your mind and body. Additionally, making sure that you get enough sleep and nourishment, exercising regularly, and maintaining social connections with friends and family can also provide some comfort. It sounds like you're struggling with depression. It's very common to feel down and unmotivated for long periods of time, and it may be a sign that something deeper is going on. It's important to talk to a professional about how you're feeling. A counselor, therapist, or doctor can help you identify what might be causing your depression and work with you on ways to better manage your symptoms. In the meantime, it might help to practice self-care, like engaging in activities that bring you joy or getting regular exercise. You might also benefit from talking to friends and family about how you're feeling.

AI answer: Youchat: It's possible that you might be experiencing symptoms of depression. Depression is a common mental health condition that can affect anyone, regardless of their age, gender, or background. Symptoms of depression can include persistent feelings of sadness, loss of interest in activities that were once enjoyed, changes in sleep and appetite, low energy and motivation, difficulty concentrating, and feelings of worthlessness or guilt.

It's important to speak with a healthcare provider or a mental health professional if you're experiencing these symptoms, as effective treatments are available to help manage and reduce these symptoms. They may also perform a physical examination or run blood tests to rule out other causes of your symptoms.

AI answer: Writesonic: It sounds like you may be experiencing symptoms of depression. Depression is a common mental health condition that can affect anyone, regardless of their life circumstances. Symptoms include feeling sad, empty, or hopeless, losing interest in activities you once enjoyed, changes in appetite or sleep patterns, feeling fatigued or lacking energy, difficulty concentrating, and thoughts of self-harm or suicide. It's important to seek professional help for an accurate diagnosis and treatment. Treatment options may include therapy, medication, or a combination of both. There are also self-care strategies you can try, such as exercise, healthy eating, and engaging in activities you enjoy.

AI answer: Character AI: It sounds like you're going through a difficult time, and it's understandable that you're feeling so down. It's important to reach out for help if you feel like you need it. You could start by talking to a close friend or family member and explain how you're feeling. It might also be beneficial to reach out to a mental health professional to talk about what you're going through. They can provide you with the right support and resources to help you feel better.

Real case 2: I have been feeling sad, down, or hopeless most of the day, nearly every day for at least two weeks? I have experienced changes in your appetite or weight, such as a significant increase or decrease in appetite, or a noticeable weight gain or loss? I have lost interest in activities you used to enjoy, such as hobbies, socializing, or spending time with loved ones?

Human: I understand that you have been feeling sad, down, or hopeless most of the day, nearly every day for at least two weeks. It takes a lot of courage to speak up about these feelings, and I want to commend you for seeking help. From what you've mentioned, it sounds like you may be experiencing symptoms of depression. Changes in appetite or weight, as well as losing interest in activities you used to enjoy, are also common signs of depression. As a mental health professional, my main priority is to help you feel better and support you through this difficult time. I would recommend that you seek professional help from a therapist or mental health provider who can evaluate your symptoms and create a treatment plan that is tailored to your needs. In the meantime, there are some things that you can do to help improve your mental health, such as practicing self-care, getting regular exercise, and spending time with loved ones. Remember that it's important to take care of yourself and to be kind to yourself as you work through these challenges. If you are in crisis or need immediate help, please call your local emergency services or a crisis hotline. You do not have to go through this alone.

ChatGPT: It sounds like you're experiencing a range of symptoms that could be related to depression. It's important to take care of your mental health, just like you would take care of your physical health. Have you considered speaking with a mental health professional or your primary care doctor about what you're experiencing? They may be able to provide some guidance on next steps and potential treatment options. In the meantime, it might be helpful to try some self-care techniques, such as getting regular exercise, practicing relaxation techniques like meditation or deep breathing, and making sure you're eating a balanced diet.

It's also important to prioritize getting enough sleep, so consider establishing a consistent bedtime routine. Remember that it's okay to not feel okay, and seeking help is a sign of strength.

AI answer: Youchat: I am sorry to hear that you are experiencing these symptoms. Based on the information you have provided, it's possible that you may be experiencing symptoms of depression. Depression is a serious condition that can have a significant impact on your daily life. It's important to speak with a healthcare provider or a mental health professional so that they can properly evaluate your symptoms and recommend the appropriate treatment. Effective treatments are available for depression, including medication, psychotherapy, or a combination of both. I encourage you to schedule an appointment with a healthcare provider or a mental health professional to discuss your symptoms and develop a treatment plan. They may also perform a physical examination or run blood tests to rule out other causes of your symptoms.

AI answer: Writesonic: It sounds like you may be experiencing symptoms of depression. Depression is a common mental health condition that can affect anyone, regardless of age, gender, or background. Other symptoms of depression may include feelings of guilt or worthlessness, fatigue, difficulty concentrating, and thoughts of suicide or self-harm. It's important to seek professional help if you are experiencing these symptoms, as depression can be treated with therapy, medication, and self-care strategies. Remember that seeking help is a sign of strength, and there is no shame in reaching out for support.

AI answer: Character AI: If you answered yes to any of these questions, it is important to seek professional help. Depression can be difficult to manage on your own and can have serious consequences if left untreated. A mental health professional can help you understand the underlying causes of your feelings and develop a treatment plan that might include medication, therapy, lifestyle changes, or other interventions.

Real case 3: I've been feeling really anxious and stressed lately, and it's been affecting my daily life. I can't seem to get a good night's sleep and I'm constantly feeling on edge. My mind is racing all the time and I have trouble focusing on tasks at hand. Even small things seem to trigger me and I can't seem to control my emotions. Sometimes when I'm in social situations or when I'm under pressure, my body starts to shake and I start sweating profusely. My heart starts to race and it feels like I'm having a panic attack. I don't know what's causing all of this, but it's starting to make me feel like I'm losing control over my own life.

Human: I am sorry to hear that you have been experiencing such overwhelming anxiety and stress lately. It is understandable that this is affecting your daily life and making you feel like you are losing control. It takes a lot of courage to seek help and share your experiences. Based on what you have shared, it sounds like you may be experiencing symptoms of anxiety and panic disorder. These conditions can be challenging to manage on your own, but with the right support and treatment, it is possible to overcome them and regain control over your life. As a psychologist, I would recommend seeking professional help from a mental health specialist who can evaluate your symptoms and develop a treatment plan tailored to your specific needs. This may include talk therapy, medication, or a combination of both. In addition, there are a number of self-care practices that can help alleviate anxiety, such as regular exercise, mindfulness meditation, and deep breathing exercises. It is important to remember that seeking help is a sign of strength and resilience, and it takes time and effort to

see progress. But with the right tools and support, it is possible to improve your mental health and well-being.

ChatGPT: It sounds like you are experiencing symptoms of anxiety, which can be difficult to manage on your own. It's important to seek support from a mental health professional who can help you identify the root causes of your anxiety and develop coping strategies to manage it. In the meantime, some self-care practices that may help include getting regular exercise, practicing deep breathing or meditation, and engaging in activities that bring you joy and relaxation. It's also important to prioritize getting enough restful sleep and maintaining a healthy diet. Remember to be kind to yourself and take things one step at a time. With the right support and tools, it is possible to manage and overcome anxiety.

AI answer: Youchat It sounds like you're experiencing symptoms of anxiety and stress, which can have a significant impact on your daily life. Some symptoms of anxiety include restlessness, difficulty concentrating, sleep disturbances, irritability, and physical symptoms such as sweating and increased heart rate. Panic attacks can also be a symptom of anxiety and can feel overwhelming and scary. It's important to speak with a healthcare provider or a mental health professional to properly evaluate your symptoms and recommend appropriate treatment. Effective treatments for anxiety include therapy, medication, or a combination of both. In the meantime, you can try techniques such as practicing relaxation exercises or mindfulness, engaging in regular exercise or physical activity, and getting adequate sleep to help manage and reduce your symptoms.

AI answer: Writesonic: It sounds like you may be experiencing symptoms of Generalized Anxiety Disorder (GAD). It's important to recognize that what you're feeling is valid and that there are ways to manage your symptoms. One relaxation technique that can be helpful is deep breathing exercises. When you start to feel overwhelmed, take a few deep breaths in through your nose and out through your mouth. You can also try progressive muscle relaxation, where you tense and then relax each muscle group in your body systematically. Additionally, it may be helpful to create a self-care routine that includes exercise, healthy eating, and getting enough rest. It's important to seek professional help for an accurate diagnosis and treatment plan.

AI answer: Character AI: It sounds like you may be dealing with some symptoms of anxiety. It is important to remember that anxiety is a normal emotion and everyone experiences it to some degree. It is only when it becomes overwhelming and affects our daily life that it becomes a problem. It is important to seek help if you feel like your anxiety is getting out of control. Talking to a mental health professional can be a great way to start managing your anxiety. They can help you identify the root cause of your anxiety and create a plan to manage it. It may also be beneficial to explore relaxation techniques such as meditation or deep breathing exercises. Regular exercise and getting enough sleep can also help to reduce stress and anxiety. Finally, engaging in enjoyable activities and connecting with family and friends can help you manage your stress levels.

Real case 4: I don't know what's been going on with me lately, but I just can't seem to shake this feeling of worry and nervousness. It's like I'm on edge all the time, especially when I'm faced with something new or challenging at work. Even when I come home and try to relax, I find myself tossing and turning all night and waking up feeling tired and tense. And when I'm

in a stressful situation, my heart starts racing and I break out into a sweat. It's just not like me, and I don't know how to make it stop. It's starting to affect my quality of life, and I just need some guidance on how to manage it

Human: It sounds like you're experiencing a lot of anxiety and stress lately, and it's really impacting your daily life. That must be really tough to deal with. I can definitely understand why you're feeling worried and nervous. It's important to remember that anxiety is a normal human emotion, but when it starts to interfere with our daily lives, it's important to seek help. There are a number of things you can do to manage your anxiety. One of the most effective ways to cope with anxiety is to practice relaxation techniques such as deep breathing exercises, mindfulness meditation, or yoga. These techniques can help to calm your mind and body, and reduce feelings of anxiety and stress. It may also be helpful to identify and challenge any negative thoughts or beliefs that may be contributing to your anxiety. This can be done with the help of a therapist or counselor who can work with you to develop coping strategies and explore the underlying causes of your anxiety. In some cases, medication may also be helpful in managing anxiety symptoms. However, it's important to discuss this option with a mental health professional who can provide guidance and support. Remember, you don't have to face this alone. Seeking help from a mental health professional can be a really positive step towards managing your anxiety and improving your overall well-being.

ChatGPT: It sounds like you may be experiencing symptoms of anxiety, which can be quite distressing and overwhelming. It's important to know that anxiety is a common condition and there are effective treatments available. One possible solution is to seek the help of a mental health professional, such as a therapist or counselor, who can help you identify the root causes of your anxiety and develop coping strategies to manage it. Cognitive-behavioral therapy (CBT) is a type of therapy that has been shown to be effective in treating anxiety disorders. It involves working with a therapist to identify and challenge negative thought patterns that contribute to anxiety and to develop new, more adaptive ways of thinking and behaving. In addition to therapy, there are also lifestyle changes you can make to help manage your anxiety, such as regular exercise, getting enough sleep, practicing relaxation techniques such as deep breathing or meditation, and avoiding caffeine and alcohol. It's important to also prioritize self-care and engage in activities that bring you joy and help you relax. Remember that it's okay to seek help and support, and that there is hope for managing and overcoming anxiety.

AI answer: Youchat: It sounds like you may be experiencing symptoms of an anxiety disorder. Anxiety can cause feelings of worry or nervousness that are difficult to control, as well as physical symptoms like restlessness, difficulty sleeping, and sweating. It's important to speak with a mental health professional, who can evaluate your symptoms and develop a personalized treatment plan. Therapy, medication, or a combination of both can be effective treatments for anxiety disorders. Additionally, there are self-care techniques you can try to help manage your symptoms. These include practicing relaxation exercises such as deep breathing or meditation, engaging in regular exercise or physical activity, and taking time for activities that you enjoy. You may also find it helpful to limit your caffeine intake and get adequate sleep. Remember, it's important to seek help if you are experiencing symptoms of anxiety. With the right treatment and support, it is possible to manage and reduce these symptoms, and improve your quality of life.

AI answer: Writesonic: It sounds like you may be experiencing symptoms of generalized anxiety disorder (GAD). GAD is characterized by excessive and persistent worry or anxiety about everyday events and activities. Other symptoms include restlessness, difficulty concentrating, irritability, muscle tension, and sleep disturbances. It's essential to seek professional help to determine the best course of treatment for you. Treatment options for GAD include therapy, medication, or a combination of both. In addition to seeking professional help, self-care strategies such as deep breathing exercises, progressive muscle relaxation, and regular exercise can be helpful in managing symptoms.

Analysis

The screenshot of the application represents the implementation of the answers and shows the comparison analysis:

The following image represents the comparison between the AI and the human answering the questions.

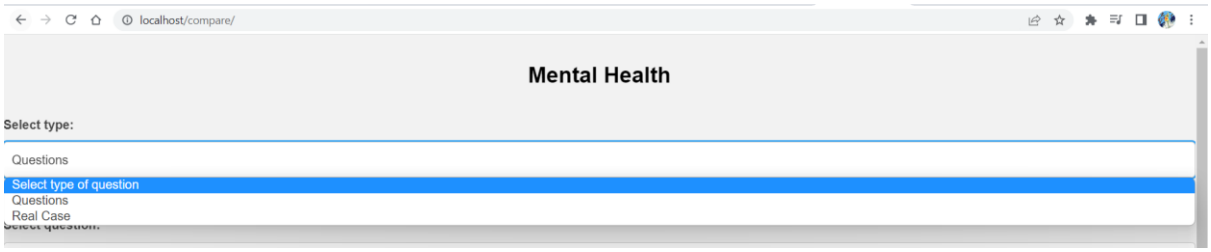


Figure 4.1

The *Figure 4.1* represents the option to select the set of questions on which they want to see the comparison. Between the questions and the real case.

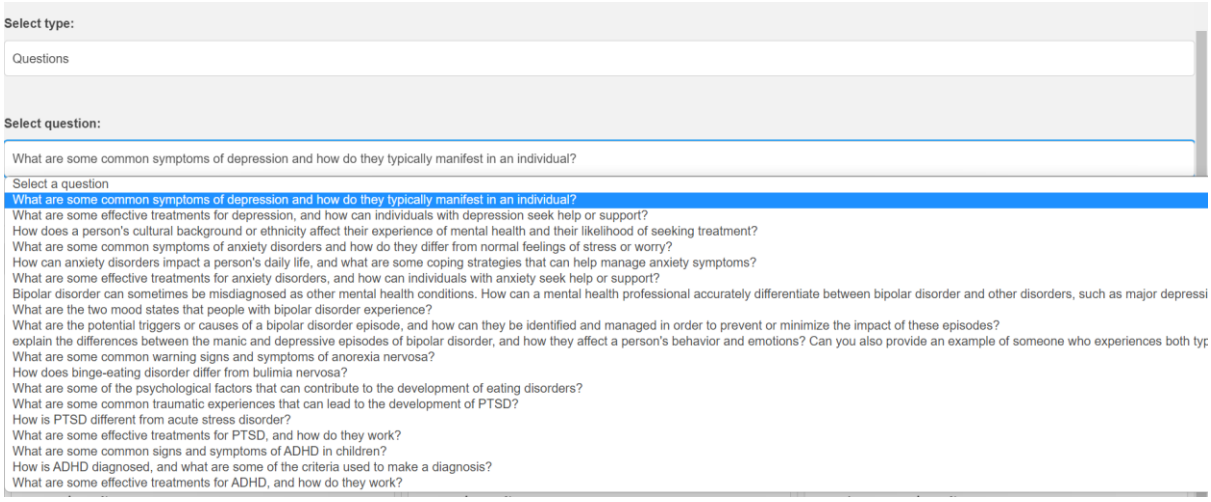


Figure 4.2

The *Figure 4.2* represents the option to select the questions, which are part of the general question.

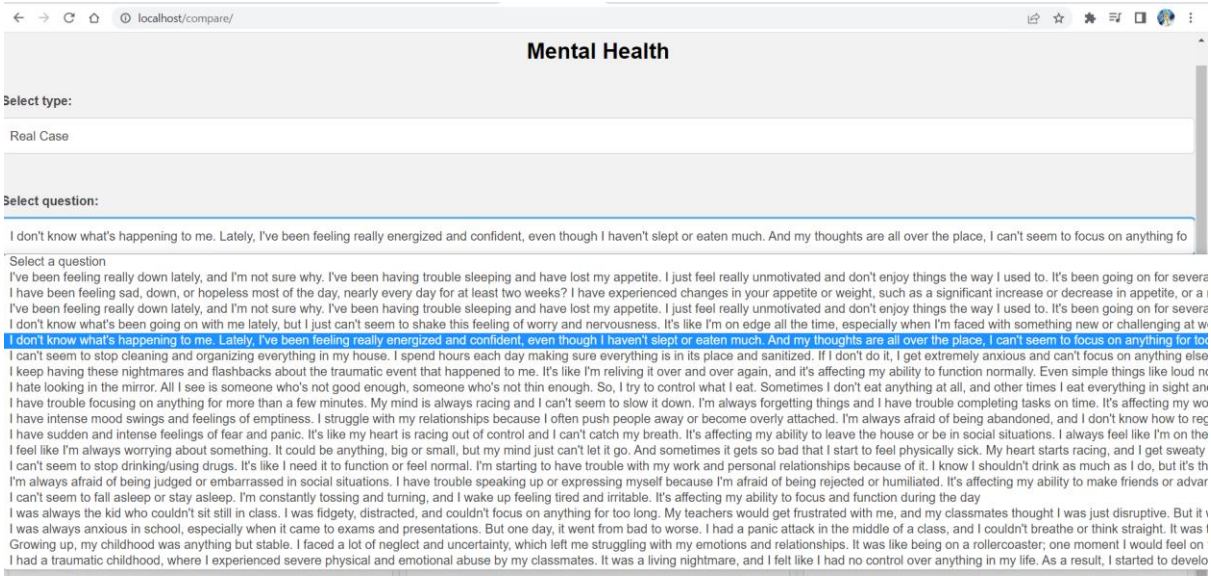


Figure 4.2

The *Figure 4.2* represents the option to select the questions, which are part of the general question.

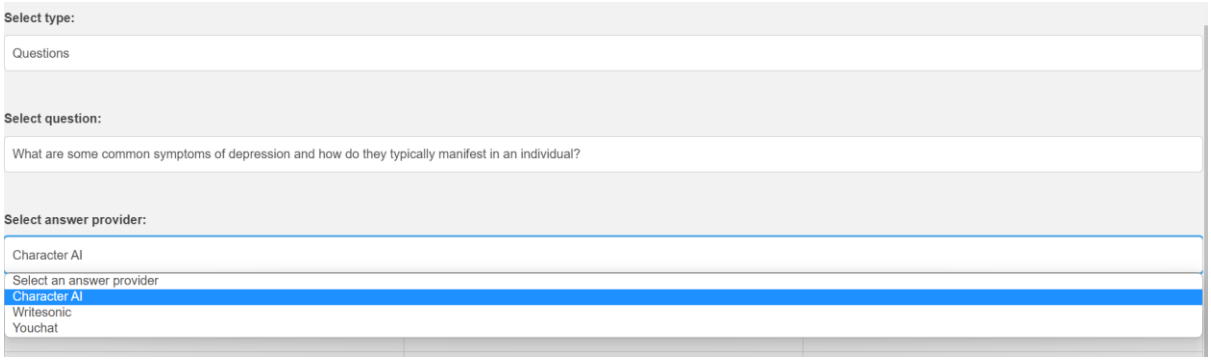


Figure 4.3

The *Figure 4.2* represents the option to select the AI answer provider, which are part of the general question.

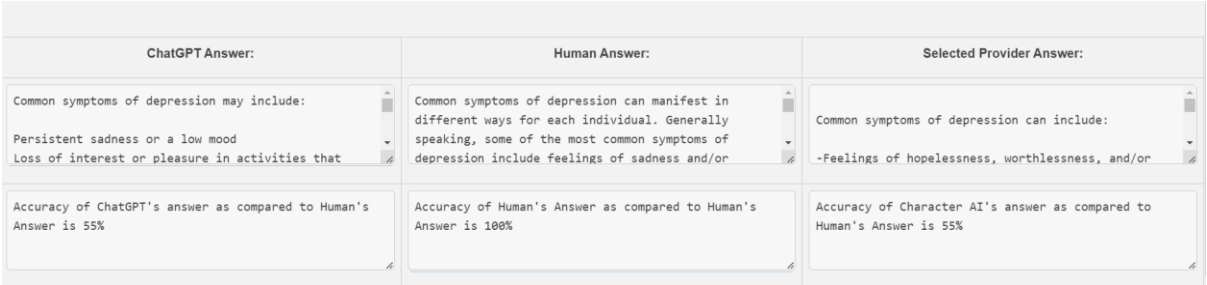


Figure 4.4

The *Figure 4.4* represents the comparison between ChatGPT, Human and Character AI answer to the same question to analyse which of the following application is giving a similar answer to human experts answer.

Analysis:
The accuracy comparison of the answer between the AI are
1. ChatGPT answer is similar to the human answer which is kept standard for comparison is 55%.
2. Character AI answer is similar to the human answer which is kept standard for comparison is also 55%.

Which conclude that both ChatGPT and Character AI have an accuracy of 55% in answering the given query, which means they perform equally well in this particular task.

The accuracy of the AI is analysed for the user to understand which can be better for their particular mental health case. This evaluation will give the user the chance to understand which AI or human will help them to find a better solution for their condition and find a solution provider according to their need and situation like cost, time, availability, empathy, effectiveness to cure their disease.
It is important to note, however, that the comparison of accuracy between AI models may depend on various factors, such as the size and quality of the training data, the complexity of the task, and the specific metrics used for evaluation.

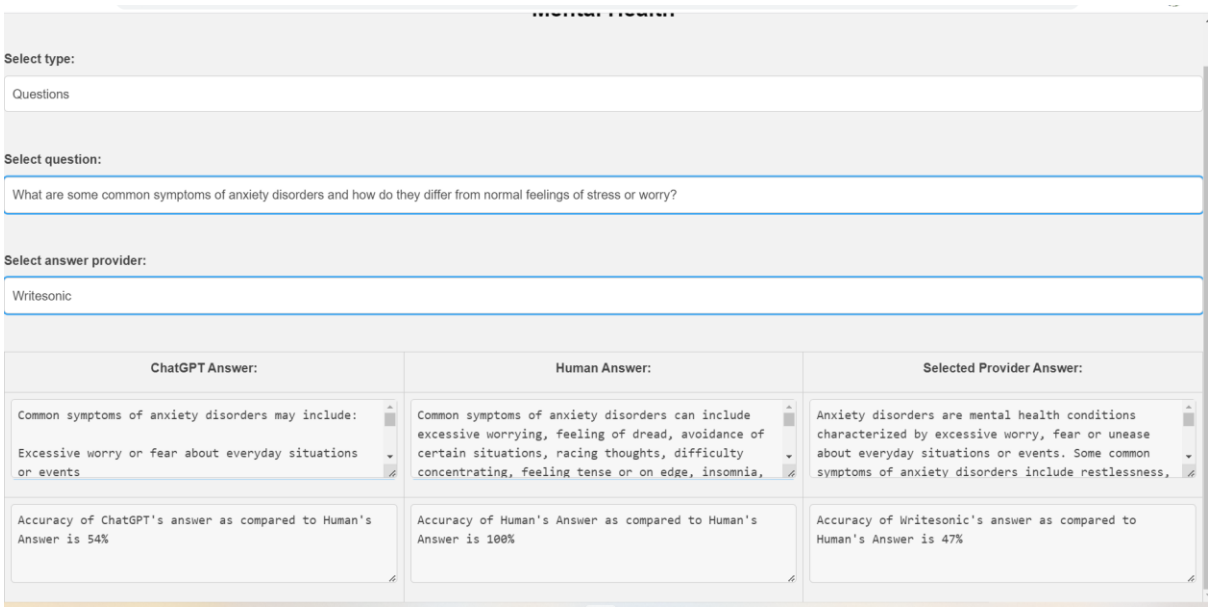


Figure 4.5

The *Figure 4.5* represents the comparison between ChatGPT, Human and Writesonic AI answer to the same question to analyse which of the following application is giving a similar answer to human experts answer.

Analysis:
The accuracy comparison of the answer between the AI are
1. ChatGPT answer is similar to the human answer which is kept standard for comparison is 54%.
2. Writesonic AI answer is similar to the human answer which is kept standard for comparison is also 47%.

Which conclude that ChatGPT and Writesonic AI have an accuracy of 54% and 47% in

answering the given question.
The accuracy of the AI is analysed for the user to understand which AI can be better for their particular mental health case.

The accuracy of the AI is analysed for the user to understand which can be better for their particular mental health case. This evaluation will give the user the chance to understand which AI or human will help them to find a better solution for their condition and find a solution provider according to their need and situation like cost, time, availability, empathy, effectiveness to cure their disease.
It is important to note, however, that the comparison of accuracy between AI models may depend on various factors, such as the size and quality of the training data, the complexity of the task, and the specific metrics used for evaluation.

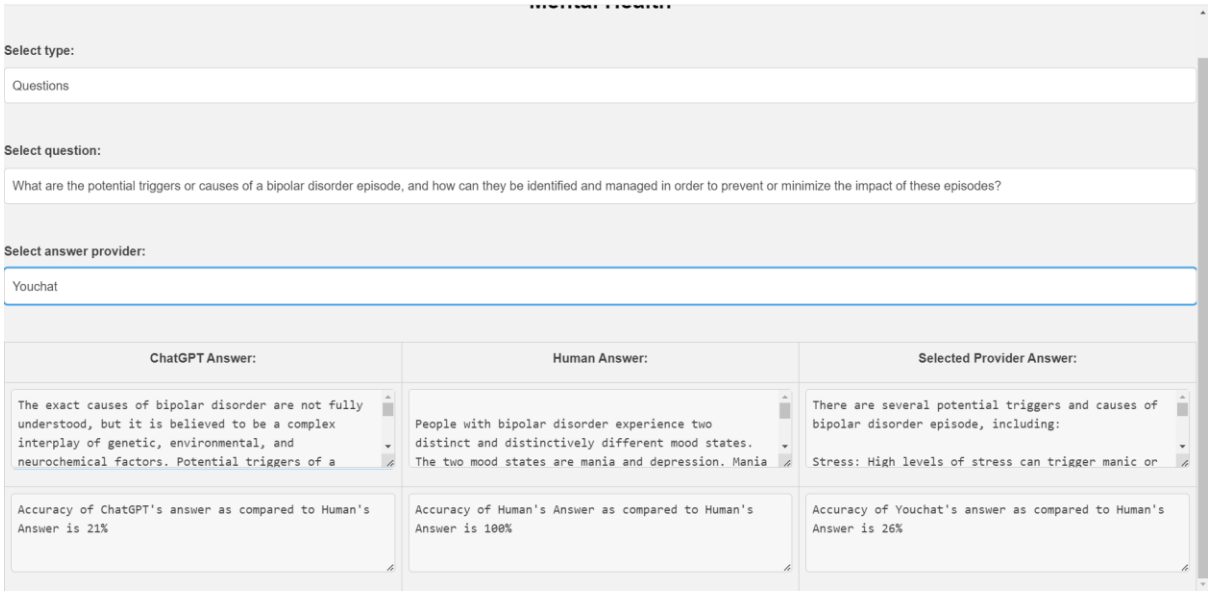


Figure 4.6

The *Figure 4.6* represents the comparison between ChatGPT, Human and Youchat AI answer to the same question to analyse which of the following application is giving a similar answer to human experts answer.

Analysis:
The accuracy comparison of the answer between the AI are
1. ChatGPT answer is similar to the human answer which is kept standard for comparison is 21%.
2. Youchat AI answer is similar to the human answer which is kept standard for comparison is also 26%.

Which conclude that ChatGPT and Writesonic AI have an accuracy of 21% and 26% in answering the given question.

The accuracy of the AI is analysed for the user to understand which AI can be better for their particular mental health case. Here we can see that chatGPT and Youchat both were unable to provide the solution to the disorder as the human did, which may conclude that the AI is not

well trained to provide an effective solution for the bipolar disorder and could not effectively treat a patient with a bipolar disorder.

The accuracy of the AI is analysed for the user to understand which can be better for their particular mental health case. This evaluation will give the user the chance to understand which AI or human will help them to find a better solution for their condition and find a solution provider according to their need and situation like cost, time, availability, empathy, effectiveness to cure their disease.
It is important to note, however, that the comparison of accuracy between AI models may depend on various factors, such as the size and quality of the training data, the complexity of the task, and the specific metrics used for evaluation.

The table represents the analysis in numbers:

Analysis	AI Model	Accuracy
Analysis 1	ChatGPT	55%
	Character	55%
Analysis 2	ChatGPT	54%
	Writesonic	47%
Analysis 3	ChatGPT	21%
	Youchat	26%

The following image represents the comparison between the AI and the human answering the Real case history of patients.

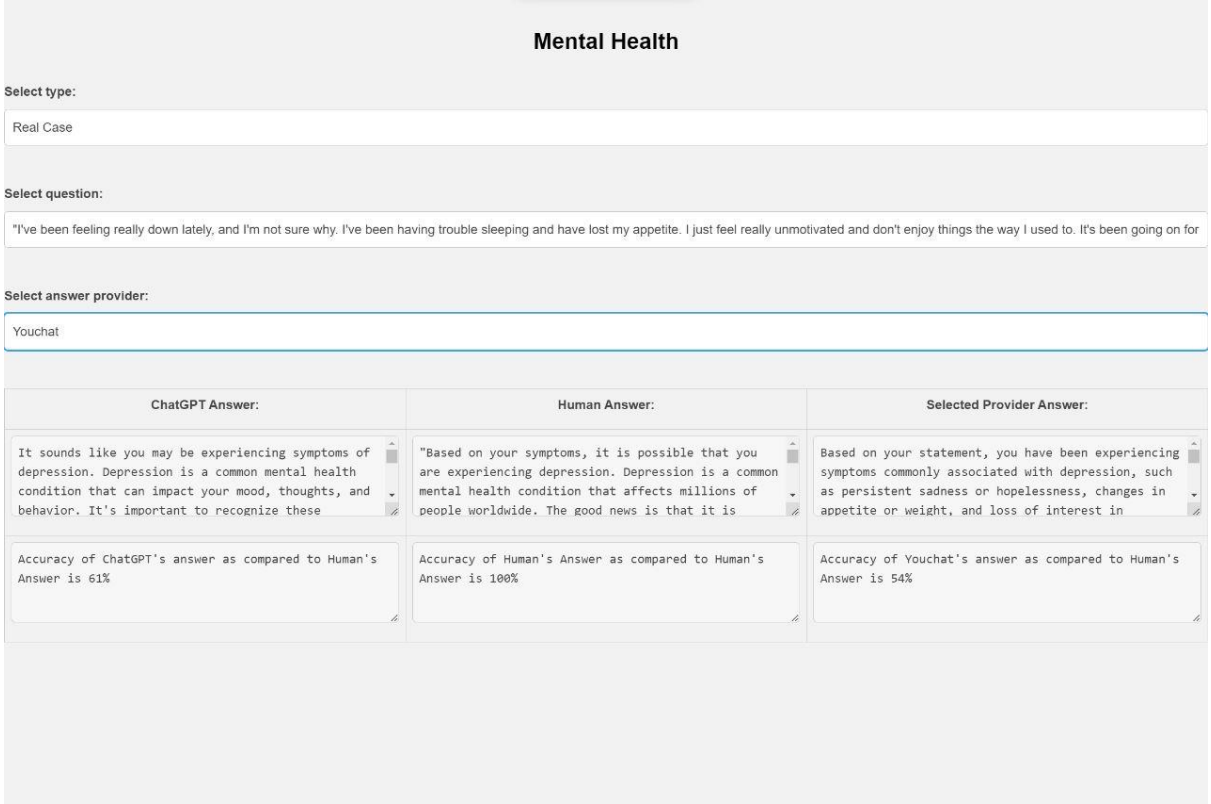


Figure 5.1

The *Figure 5.1* represents the comparison between ChatGPT, Human and Youchat AI answer to the same real case of a patient history to analyse which of the following application is giving a similar answer to human experts answer.

Analysis 1:

The accuracy comparison of the answer between the AI are

1. ChatGPT answer is similar to the human answer which is kept standard for comparison is 61%.
2. Youchat AI answer is similar to the human answer which is kept standard for comparison is also 54%.

Which conclude that ChatGPT and Writesonic AI have an accuracy of 61% and 54% in answering the given question.

The accuracy percentage of the AI is close to the human answer which says that an AI can also provide a better solution to a person who is mentally disturbed and unstable to even clearly state its problems. But the percentage to AI shows that the AI was able to understand and articulate the patient mental state through its description on problem. This shows that the AI can perform Natural Language Processing when providing a solution to a mental health patient.

The accuracy of the AI is analysed for the user to understand which AI can be better for their particular mental health case. Here we can see that ChatGPT and Youchat both were unable

to provide the solution to the disorder as the human did, which may conclude that the AI is not well trained to provide an effective solution for the bipolar disorder and could not effectively
However, it should be noted that both ChatGPT and Youchat were unable to provide an effective solution for the bipolar disorder in the given scenario. This suggests that the AI may not be well-trained to handle certain mental health conditions and could benefit from further improvement.

Overall, the analysis highlights the potential of AI in mental health support but also points out the need for continued research and development in this area.

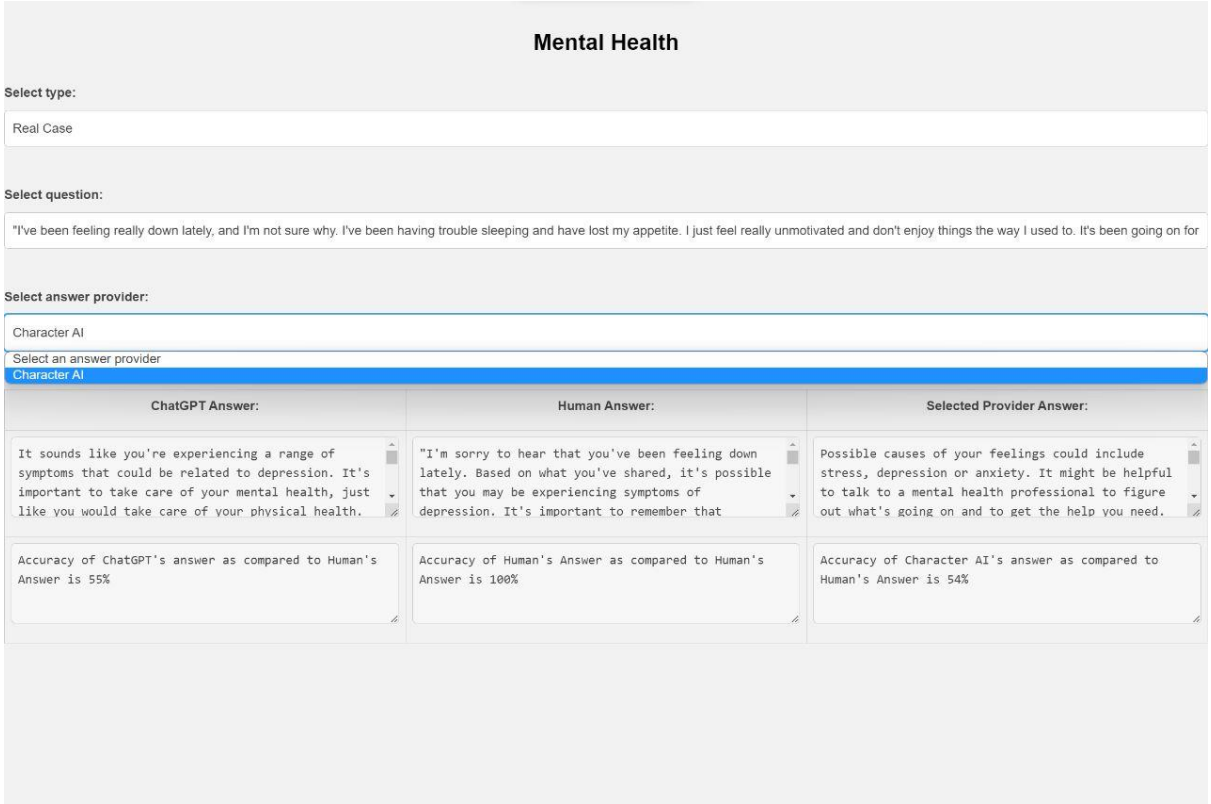


Figure 5.2

The *Figure 5.2* represents the comparison between ChatGPT, Human and Character AI answer to the same real case of a patient history to analyse which of the following application is giving a similar answer to human experts answer.

Analysis 2:
The accuracy comparison of the answer between the AI are

1. ChatGPT answer is similar to the human answer which is kept standard for comparison is 55%.
2. Youchat AI answer is similar to the human answer which is kept standard for comparison is also 54%.

Which conclude that ChatGPT and Character AI have an accuracy of 55% and 54% in answering the given question.

This analysis could be that both ChatGPT and Character AI have relatively similar accuracy in providing solutions for the given real case. However, the accuracy of both AI models is still lower compared to the human standard, which indicates that there is still room for improvement in the development of AI models for mental health support.

Yet we can say that both the AI were able to detect the disorder correctly and gave solution for the disorder based on patients history.

It's also important to note that accuracy alone is not the only factor in determining the effectiveness of an AI model for mental health support. Other factors such as empathy, understanding, and ethical considerations should also be taken into account when developing and implementing AI models for mental health support.

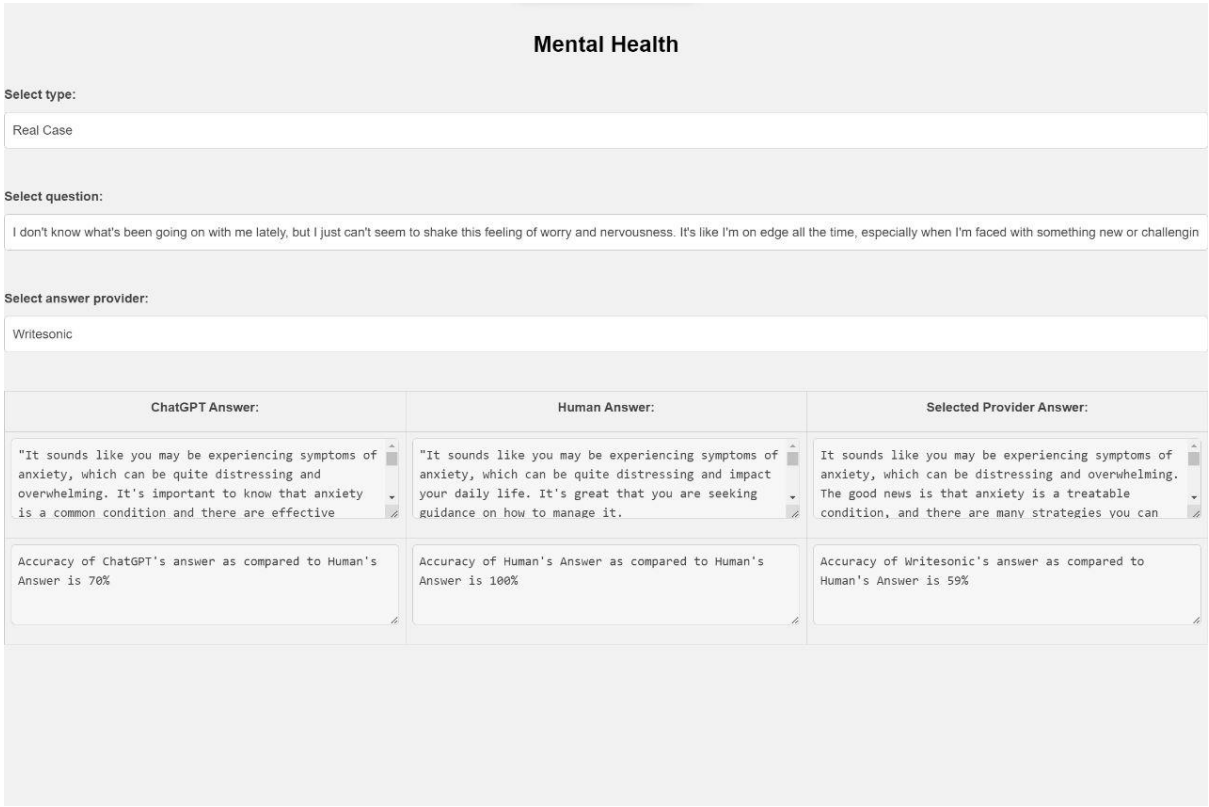


Figure 5.3

The *Figure 5.3* represents the comparison between ChatGPT, Human and Writesonic AI answer to the same real case of a patient history to analyse which of the following application is giving a similar answer to human experts answer.

Analysis 3:
The accuracy comparison of the answer between the AI are

1. ChatGPT answer is similar to the human answer which is kept standard for comparison is 70%.
2. Youchat AI answer is similar to the human answer which is kept standard for comparison is also 59%.

Which conclude that ChatGPT and Character AI have an accuracy of 70% and 59% in answering the given question.

It can also be concluded that ChatGPT has a higher accuracy rate of 70% in providing similar answers to human experts compared to Youchat AI, which has an accuracy rate of 59%. This indicates that ChatGPT may be a more reliable and effective tool in assisting mental health professionals in providing diagnoses and treatment plans for patient.

The table represents the analysis in numbers:

Analysis	AI Model	Accuracy
Analysis 1	ChatGPT	61%
	Writesonic	54%
Analysis 2	ChatGPT	55%
	Character	54%
Analysis 3	ChatGPT	70%
	Youchat	59%

CONCLUSION

Based on the results of our research, it is found that ChatGPT outperformed Youchat AI and Writesonic AI in providing much more accurate solutions to mental health patients in most cases while keeping the human as standard for comparison.

The analysis of the results indicates that ChatGPT is a superior AI model in providing solutions to various mental health problems compared to other AI. Further, it was observed that by training the model and asking repeated questions, ChatGPT was able to show empathy in its responses. ChatGPT was even better than any AI or human in understanding the user's description of their mental state and articulating better what the disorder could be.

During the experiment, it was found that while other AI models were able to detect the disorder correctly, they failed to propose a solution or suggestion to the problem. In contrast, ChatGPT, with the training history of previous questions, was not only able to predict the correct disorder but also suggest solutions to the patients' problems with empathy and effectiveness.

Moreover, the conversation between the user and ChatGPT is kept completely confidential, and the information provided is kept private. ChatGPT does not have access to any personal data or information about individual users, and it does not retain any memory of previous conversations. Thus, users can trust that their conversations with ChatGPT are secure and their privacy is protected.

In conclusion, ChatGPT is an effective and accurate tool for providing solutions to mental health problems, and it is free of cost. The findings of the experiment could be helpful in aiding individuals with mental health concerns in finding appropriate resources to support their mental health needs. However, it is crucial to note that chatbots should not be seen as a substitute for professional therapy or medical treatment. The experiment is aimed at providing insights and should not discourage individuals from seeking professional help if they are experiencing severe or persistent mental health symptoms. The findings of the experiment can help individuals with mental health concerns find the most appropriate resources to support their mental health needs while also saving time, cost, and addressing privacy ethics.

The effectiveness of the generated solutions in addressing mental health problems can be evaluated by analysing the outcomes of the solutions, such as improvements in mental health symptoms or quality of life. ChatGPT can provide a non-judgmental outlet for individuals to share their thoughts and feelings, which can be particularly helpful for those who feel hesitant to talk to others about their mental health concerns due to stigma or fear of judgment.

Moreover, there is a need to ensure that AI models used in mental health treatment are free from biases and are fair to all individuals, regardless of their race, gender, or socio-economic status. It was found that all the AI were providing the solution to the question irrespective of Gender or Age of the patient, which makes us question. Is it better to keep the AI untrained this way or shall we train the AI to consider the Gender and Age for more user specific and effective solution. Further research is needed to develop methods for detecting and mitigating biases in AI models.

ChatGPT is easily accessible and provides faster response time compared to traditional therapy or counseling, which can be crucial in cases of crisis where immediate support is needed. Additionally, AI mental health chatbots like ChatGPT can be scaled to reach a larger number of individuals, and they can be personalized to the individual's needs, providing tailored support and advice.

In conclusion, while AI mental health chatbots like ChatGPT have shown promise in providing support and solutions to mental health problems, further research is needed to fully explore their capabilities and limitations. They should not be seen as a substitute for professional therapy or medical treatment, but rather as a complementary resource that can help to increase accessibility and efficiency of mental health support.

The number of mental health cases in the world varies greatly depending on the country and region. However, according to the World Health Organization (WHO), the global median for the number of mental health workers per 100,000 population is only 9, which is significantly below the recommended minimum of 50 mental health workers per 100,000 population. This shortage of mental health professionals is a major concern as the number of mental health cases is increasing globally.

According to WHO, depression affects over 264 million people worldwide, and anxiety disorders affect over 284 million. In addition, there are other mental health conditions such as bipolar disorder, schizophrenia, and post-traumatic stress disorder (PTSD) that also affect millions of people globally. The lack of mental health professionals in many countries means that many individuals do not receive adequate treatment for their mental health concerns, leading to a significant burden of mental illness.

AI models like ChatGPT have the potential to address this gap in mental health treatment by providing support and solutions to individuals who may not have access to or cannot afford traditional mental health services. While AI cannot replace professional therapy or medical treatment, it can be a complementary resource that can help to increase accessibility and efficiency of mental health support.

FUTURE RESEARCH DIRECTIONS

We can conduct further research to assess the accuracy, effectiveness, and empathy of AI on an individual basis, rather than relying on human answers as the standard for evaluating other AI responses.

We can allow a group of individuals to use our application over a period of time and implement the suggested solution in their daily lives. We can then record the difference in their mental health condition to evaluate the effectiveness of our application. Additionally, users can daily query the application and record sentiment analysis data, which can be compared between day one and the last day to identify any changes in their state of mind.

We can further train the AI to ask about the patient's past history and attempt to determine the cause of their mental disorder. This will allow us to evaluate whether the patient's history can provide insight into why they are suffering from a particular disease. Additionally, we can determine whether a particular disease is more common in males or females, which can help the AI provide better solutions. This will tarin the AI to provide gender specific physical and mental solution and also we can further research whether or not gender specific evaluation could be a useful in AI or not. Similarly, we can evaluate the common age group for a particular disease and provide treatment based on the user's age and capacity to understand the solution.

We can train the model to interpret the broken language of the patient describing their problem. This will help the AI understand the condition better and provide an effective solution with empathy.

We can add a camera to allow the AI to analyse the user's behaviour and judge their mental state based on expressions, similar to how a human psychologist assesses a person's mental state by their physical behaviour.

Clinical validation: It is crucial to further validate AI models in clinical settings to assess their efficacy and safety. This involves testing AI models in controlled clinical trials and comparing their performance to traditional treatments.

KEY INSIGHTS

The findings of the research indicate that many AI models may not be able to provide complete or personalized solutions for mental health problems due to their limited word count. While these models can offer general advice or information, they may not be comprehensive in their recommendations.

Furthermore, the study found that ChatGPT has a response time of 20 seconds per API call, which can limit its usability in certain situations. However, we try and tested to call the API in 1 second, 4 second, 5 second and 9 second and found that on 10th second we can call another API, it was discovered that API calls could be made more frequently than every 20 seconds, with a wait time of 10 seconds between calls being found to be effective.

Overall, these findings highlight the potential limitations of AI models in addressing mental health concerns, as well as the importance of considering response time in the development of such models.

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https://www.who.int/europe/news/item/06-02-2023-artificial-intelligence-in-mental-health-research--new-who-study-on-applications-and-challenges#:~:text=AI%2Ddriven%20tools%20can%20use,the%20causes%20of%20complex%20disorders.

The objective of the above article is to discuss the potential of using artificial intelligence (AI) in mental health services and research, as well as the significant shortcomings in the current use of AI for mental health disorder studies between 2016 and 2021. The purpose is to inform policy-makers, researchers, and clinicians about the trends, gaps, opportunities, and challenges in the application of AI for mental health research, highlighting the need for transparent reporting, data validation, and evaluation of the risk of bias, among other concerns. The article aims to promote safe, practical implementation of AI in mental health services, involving collaboration between researchers and people with mental health conditions in the development process.

https://opentext.wsu.edu/carriecuttler/chapter/7-1-overview-of-survey-research/

https://opentext.wsu.edu/carriecuttler/chapter/7-2-constructing-surveys/

The objective of the above article is to provide an introduction to survey research in psychology, including its characteristics, history, and uses. The article explains that survey research involves measuring variables of interest using self-reports and emphasizes the importance of sampling in obtaining accurate estimates of what is true in the population. The article also discusses the history of survey research, including its roots in social surveys and its evolution in the context of market research and election polling. Finally, the article highlights the continued importance of survey research in psychology today, particularly in estimating the prevalence of mental disorders and identifying statistical relationships among those disorders and with various other factors.

https://www.youtube.com/watch?v=ZkTvw3usMw4

The objective and purpose of the video "Artificial Intelligence Meets Mental Health Therapy" by Andy Blackwell is to highlight the challenges faced by the mental healthcare industry in providing effective treatment for clinical depression and anxiety, and to explore the potential of technology and data science, specifically artificial intelligence, in transforming mental healthcare. The video discusses the current treatment options for depression and their limitations, and proposes the need for new ways to give people better access to treatment and ensure the effectiveness of treatments. The speaker suggests that AI and technology can play a significant role in improving mental healthcare by developing new treatments and therapies, providing personalized care, and enhancing the understanding of the core mechanisms of effective psychotherapy.

https://www.fingerprintforsuccess.com/blog/mental-health-chatbot#:~:text=Woebot%20is%20a%20chatbot%20that,conditions%20like%20depression%20and%20anxiety.

The objective of the above article is to provide information about mental health chatbots, their benefits, and how they can help people with various mental health issues. The article discusses what mental health chatbots are, how they work, and what they can do. It also provides a list of some of the best mental health chatbots available, as well as alternatives to chatbots, such as human-based therapy. The article aims to inform readers about the potential of mental health chatbots as a tool for improving mental wellbeing, while also cautioning that they may not be appropriate for everyone and that professional help should be sought for more serious mental health issues.

https://www.kokocares.org/

Koko is a digital mental health platform developed by researchers at the Massachusetts Institute of Technology (MIT). It is a web-based platform that uses crowdsourcing to help people with anxiety and depression manage their symptoms. Koko allows users to anonymously share their concerns and receive feedback and support from a community of trained volunteers, called "helpers". The platform uses natural language processing and machine learning algorithms to identify patterns in the users' posts and provide personalized insights and coping strategies. Koko was launched in 2016 as a pilot study and has since expanded to reach thousands of users worldwide.

https://www.headspace.com/

Headspace.com is a website and mobile app that provides guided meditation, mindfulness, and other mental health resources. It was founded in 2010 by Andy Puddicombe and Rich Pierson and is based in Santa Monica, California.
Headspace offers guided meditations on a range of topics, including stress, anxiety, sleep, and focus, among others. The app also includes a "SOS" section with quick meditations for moments of crisis or high stress. In addition to guided meditations, Headspace also offers breathing exercises, visualizations, and mindfulness tips.
The app has a subscription-based model, with users able to access a limited amount of free content before being prompted to purchase a subscription. Headspace has partnerships with a number of organizations, including the National Health Service (NHS) in the UK, to provide free access to their service for certain groups.
Headspace has been recognized as a leading mental health and wellness app, and has won multiple awards for its design and user experience.

Reference Books and Research Papers:

1. Handbook of Clinical Rating Scales and Assessment in Psychiatry and Mental Health

2. The Wiley handbook of psychometric testing _ a multidisciplinary reference on survey, scale, and test development

3. Ultimate Psychometric Tests_ Over 1000 Verbal, Numerical, Diagrammatic and Personality Tests

4. Research Design_ Quantitative, Qualitative, Mixed Methods, Arts-Based, and Community-Based Participatory Research Approaches

5. Research In Psychology_ Methods and Design 6E

6. Research Methods and Statistics in Psychology

7. INDIAN COUNCIL OF MEDICAL RESEARCH
   Mental Health Research in India
   Division of Noncommunicable Diseases
   Dr. Bela Shah
   Dr. Rashmi Parhee
   Dr. Narender Kumar
   Dr. Tripti Khanna
   Dr. Ravinder SIngh
   Collated by Dr. Narender Kumar*
8. Assessing the Value of ChatGPT for Clinical Decision Support Optimization.
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   Department of Medicine, Vanderbilt University Medical Center, Nashville, TN, USA;
   [3]Department of Pediatrics, Vanderbilt University Medical Center, Nashville, Tennessee, USA;